

Consulting Project Report

Uncovering Structural Relations in the ifo Business Survey

Department of Statistics
Ludwig-Maximilians-Universität München

Jakob Freytag, Daniel Gloukhman, Dominik Pitz

Munich, August 22th, 2025



Project Parter: Dr. Klaus Wohlrabe, ifo Institute
Supervised by Dr. Fabian Scheipl, LMU

Abstract

The ifo Business Survey is a monthly poll conducted by the ifo Institute for Economic Research in Germany, gathering responses to self-assessment questions on topics such as business expectation or current situation from multiple companies across different sectors. Its most popular indicator is the ifo Business Climate Index, aggregating the business sentiment across all industries. While the index is well-established as a leading indicator, the full potential of the time series of the underlying sub-aggregates remains largely unexplored. This report accompanies the consulting project, in which we aimed to uncover potential relationships by taking a closer look at the manufacturing industry, whose business climate is closely linked to the main index. By analyzing the industry's sub-aggregates over time, we examine how their interactions with the manufacturing business climate evolve and whether certain sub-aggregates can serve as early indicators of turning points. Detecting such patterns could enhance the economic assessments accompanying the monthly release of the ifo Business Climate Index.

We begin with a comprehensive correlation analysis to map the relationships between individual industries and the main manufacturing index. These relationships are further examined by a Granger causality analysis, where we find that when formally tested, almost all individual subseries are Granger causal, but their information gain is negligible. We then use different approaches to identify subsets which seem to be especially indicative, such as an elastic net, a scoring model and clustering based on their cross-correlation function with the main index. At last, we assess, whether we can predict economic turning points of the main index, using Markov Switching models applied to the sub-aggregates. In this setting, neither individual, nor grouped series were able to meaningfully indicate the main turning points.

Contents

1	Introduction	1
1.0.1	Data	1
2	Correlation Analysis	4
2.1	Analysis Setup	4
2.1.1	Correlation Analysis Approach	4
2.1.2	Data Structure	4
2.1.3	Stationarity	5
2.1.4	Correlation Metrics	6
2.2	Correlation Distribution	7
2.2.1	Correlation Distribution by Time	7
2.2.2	Correlation Distribution by Sector	9
2.2.3	Correlation Distribution by Level	10
2.3	Consensus Analysis	11
3	Granger Causality	13
3.1	Methodology	13
3.1.1	Motivation and Definition	13
3.1.2	Testing for Granger Causality	13
3.2	Results	13
3.2.1	Instantaneous Granger Causality	13
3.2.2	Simple Granger Causality	17
3.3	Conclusion for Granger Causality	19
4	Indicative Subsets	20
4.1	Elastic Net	20
4.2	Scoring Model	20
4.3	Model Performance	21
4.3.1	Prediction Set	22
4.3.2	Early Prediction Set	23
4.3.3	Performance Comparison and Interpretation	25
4.4	Clustering based on CCF	28
4.4.1	Comparing cluster groups	28
4.4.2	Exemplary Cluster	29
5	Turning Point Indication	31
5.1	Markov-Regime-Switching	31
5.2	Classification Task	32
5.2.1	Grouped Series - Flipped	32
6	Conclusion	34
7	Contributions	36
8	Appendix	37

9 Electronic Supplement

IV

List of Figures

1	ifo Business Climate Index over the period considered in this analysis. . . .	1
2	Number of industry codes by sector and hierarchical level.	5
3	Comparison of stationarity across time windows and questions.	6
4	Comparison of median cross-correlation and distance-correlation profiles across industries.	7
5	10-year rolling window analysis (1-month step), 2000–2025.	8
6	Peak distance correlation by question, with median and inter-quantile ranges, Levels 1–3.	8
7	Distribution of peak distance correlations by sector.	9
8	Distribution of peak distance correlations by hierarchical level.	10
9	Rolling-window distribution of peak distance correlations by level.	10
10	Consensus of top-30 sets (Jaccard index) vs. union size over time.	12
11	Distribution across questions indicating instantaneous Granger causality. .	14
12	Adjusted R^2 differences (univariate).	15
13	Main index vs. top-performing industry	17
14	Distribution across questions indicating simple Granger causality.	18
15	Adjusted R^2 differences (univariate).	19
16	Prediction performance (Prediction Set).	21
17	Prediction performance of different models (Prediction Set).	23
18	Prediction performance of different models (Early Prediction Set).	25
19	Comparison of median cross-correlation profiles across industries for the Prediction Set. Yellow boxes indicate peak lags; vertical dashed line marks zero.	25
20	Comparison of median cross-correlation profiles across industries for the Early Prediction Set (lead 4–6 months).	26
21	Distribution of R^2 across 500 random 5-code subsets, with benchmark (lagged main index), Elastic Net, and Scoring Model overlaid.	26
22	Mean CCFs of the top 5 cluster averages.	29
23	Main index (blue) vs individual series in cluster 3.	29
24	Bry-Boschan Turning Points of the main index.	31
25	Turning points of group of all level 1 series with roughly composite moving characteristics according to their peak CCF lag value.	33

List of Tables

1	Overview of manufacturing sectors covered by the ifo Business Survey . . .	2
2	ifo Business Survey questions and their translations.	3
3	Top 5 industries by difference in adjusted R^2 (univariate).	16
4	Top 5 industries by difference in adjusted R^2 (multivariate).	16
5	Top 5 industries by difference in adjusted R^2 (univariate).	19
6	Industry codes selected by the Scoring Model	22
7	Industry codes selected by the Elastic Net	22
8	Prediction performance (Prediction Set)	23
9	Industry codes selected by the Scoring Model (Early Prediction Set)	24
10	Industry codes selected by the Elastic Net (Early Prediction Set)	24
11	Prediction performance (Early Prediction Set)	24
12	Top 3 random subsets (Prediction Set)	27
13	Top 3 random subsets (Early Prediction Set)	27
14	Top 5 clusters based on MSE	28
15	Individual series present in cluster 3.	30
17	Evaluation of the flipped indication setting.	33
18	Industry Codes and Titles	37

1 Introduction

The ifo Business Survey is a monthly poll conducted since 1949 by the ifo Institute for Economic Research in Germany, gathering responses from multiple firms across different sectors (?). It includes self-assessment questions on topics such as the current business situation or expectation and also more specific questions, such as production plans. The answers are aggregated hierarchically, ranging from small specific industries up to four main sectors: manufacturing industry, construction industry, service sector and trade. The survey’s most popular indicator is the ifo Business Climate Index, consisting of the the two questions on current business situation and expectation, aggregated across all industries. The index is well-established as a leading indicator, with many studies examining its predictive power for economic variables such as the national GDP. However, the underlying structures within the business survey are largely unexplored so far. In this project, we will take a closer look on the many time series underlying the ifo Business Climate Index. Rather than focusing on building a rigorously validated predictive model, this project seeks to uncover potential relationships in these time series through a descriptive analysis of historical data.

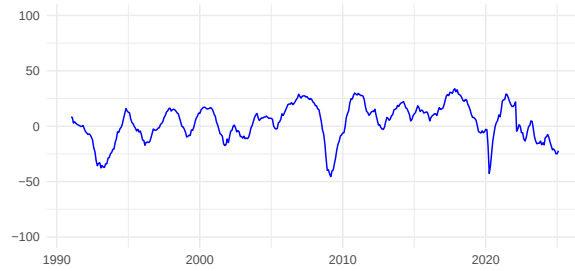


Figure 1: ifo Business Climate Index over the period considered in this analysis.

1.0.1 Data

Within the survey, we focus on the manufacturing sector, one of the main drivers of the German economy (?). In our dataset, the highest aggregation level corresponds to the entire manufacturing sector as a whole. One level below this aggregate in the hierarchical structure, there are 21 constituent industries such as the automotive industry or the machinery and equipment industry. We refer to these as level 1 aggregates. In total, five levels exist, though our analysis mainly considers level 1 and level 2 aggregates. Industry aggregates are identified by an industry code based on the official classification of economic activities of the German Federal Statistical Office (?). For example, the main aggregate is coded C0000000, with the letter C denoting the manufacturing industry and the digits indicating the aggregation level, which is the the highest level in this case, as all digits are zero. Industries at level 1 are identified by the first two digits, e.g. C2800000 for machinery and equipment. Each additional nonzero digit specifies one further level, so C2810000 corresponds to a level 2 sub-aggregate of machinery. In our study we define level 1 aggregates (codes C followed by two digits) as sectors. Table 1 provides an overview of the manufacturing sectors covered by the survey. The abbreviations for all other industry

codes can be found in the appendix 8.

Code	Sector
C1000000	Manufacture of food products
C1100000	Manufacture of beverages
C1300000	Manufacture of textiles
C1400000	Manufacture of wearing apparel
C1500000	Manufacture of leather and related products
C1600000	Manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials
C1700000	Manufacture of paper and paper products
C1800000	Printing and reproduction of recorded media
C1900000	Manufacture of coke and refined petroleum products
C2000000	Manufacture of chemicals and chemical products
C2100000	Manufacture of basic pharmaceutical products and pharmaceutical preparations
C2200000	Manufacture of rubber and plastic products
C2300000	Manufacture of other non-metallic mineral products
C2400000	Manufacture of basic metals
C2500000	Manufacture of fabricated metal products, except machinery and equipment
C2600000	Manufacture of computer, electronic and optical products
C2700000	Manufacture of electrical equipment
C2800000	Manufacture of machinery and equipment n.e.c.
C2900000	Manufacture of motor vehicles, trailers and semi-trailers
C3000000	Manufacture of other transport equipment

Table 1: Overview of manufacturing sectors covered by the ifo Business Survey

The ifo survey covers multiple questions on the current situation and expectations. For consistency, we restrict our analysis to questions and aggregates that have been included continuously from 1991 up to February 2025, the last month in our dataset. Furthermore, we only include questions that are asked on a monthly basis to avoid mixing indices of different frequencies (monthly, quarterly, or yearly). Aggregates with any missing values for a given question during this period were excluded. Table 2 lists the survey questions and their corresponding abbreviations.

Abbreviation	German Term	English Translation
KLD	Geschäftsklima	Business Climate
GUS	Geschäftslage Beurteilung	Business Situation Assessment
GES	Geschäftslage Erwartungen	Business Situation Expectations
LUS	Fertigwarenlager Beurteilung	Finished Goods Inventory Assessment
BUS	Auftragsbestand Beurteilung	Order Backlog Assessment
XUS	Auftragsbestand Beurteilung Export	Export Order Backlog Assessment
AVS	Nachfrage gegen Vormonat	Demand compared to Previous Month
BVS	Auftragsbestand gegen Vormonat	Order Backlog compared to Previous Month
QVS	Produktion gegen Vormonat	Production compared to Previous Month
PVS	Preise gegen Vormonat	Prices compared to Previous Month
QES	Produktionspläne	Production Plans
PWS	Preiserwartungen	Price Expectations
XES	Exportenerwartungen	Export Expectations
BES	Beschäftigtererwartungen	Employment Expectations
PRS	Produktivitätserwartungen	Productivity Expectations

Table 2: ifo Business Survey questions and their translations.

In our analysis we define the level 0 aggregate of the business climate question (KLD) for the manufacturing sector as the main index, which serves as the central reference point throughout the study.

2 Correlation Analysis

2.1 Analysis Setup

2.1.1 Correlation Analysis Approach

The central aim of the correlation analysis is to better understand how individual industries relate to the overall dynamics of the manufacturing sector. In particular, we address two guiding research questions. First, we ask whether there is a systematic *lead/lag structure* between industry-level indicators and the main manufacturing index. Identifying such structures would reveal which industries tend to anticipate movements in the aggregate climate and which instead follow with a delay. Second, we investigate whether it is possible to identify *stable subsets of industries* that, across different survey questions and time windows, consistently capture the evolution of the main index. The existence of such subsets would suggest that certain industries carry disproportionate informational value for monitoring the business cycle.

To answer these questions, we adopt a multi-step approach. We begin with a set of *full-sample and rolling window analyses* to estimate correlations between industry-specific indicators and the main index over time. This allows us to detect both long-term structural relationships and short-term temporal fluctuations. Next, we apply a *ranking-based consensus analysis*, which tracks how the relative importance of industries changes across questions and windows. This step provides a way to monitor stability and detect re-orderings in the correlation structure. Finally, we develop a *scoring model* that integrates information across time and questions in order to identify those industries that remain consistently close to the main index. Taken together, these steps provide a framework for moving beyond one-off correlation estimates towards a systematic evaluation of persistence, stability, and cross-question robustness.

The focus throughout is on correlations between the survey’s main manufacturing index and the industry-level indicators. All industry codes are analyzed individually, and we highlight industries whose correlations are either particularly strong or reveal distinctive lead/lag patterns. These industries are of special interest as they may function as early-warning signals or as anchors for constructing composite indicators.

2.1.2 Data Structure

The structure of the underlying data introduces important considerations for the correlation analysis. Industry codes are distributed unequally across sectors and hierarchical levels, as illustrated in Figure 2. This imbalance has non-trivial effects: rather than distorting results, the over-representation of certain sectors appears to *reinforce* the observed correlation structure. In particular, retaining the natural weighting implied by the hierarchy yields stronger and more coherent correlations between industry-level indicators and the main question than alternative approaches where correlations are first aggregated within sectors (e.g., by mean or median) and then averaged across sector values.

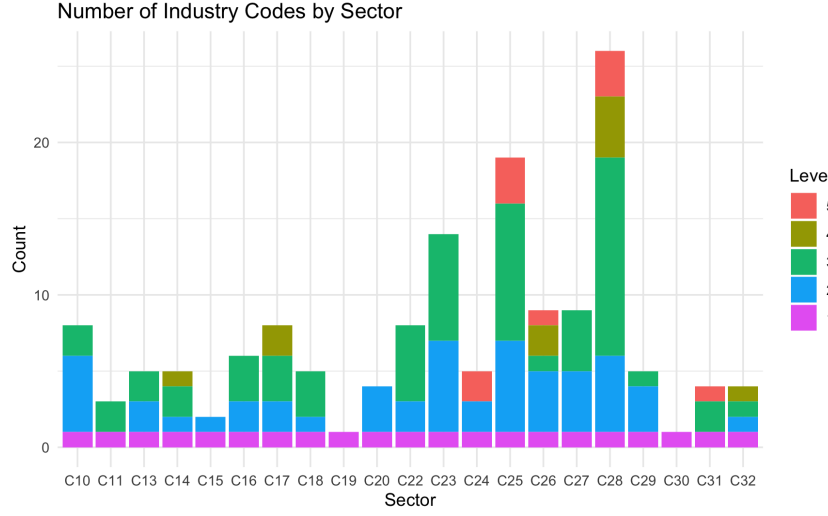


Figure 2: Number of industry codes by sector and hierarchical level.

This observation suggests that the hierarchical design of the industry codes reflects genuine structural dependencies in the data. For this reason, levels 4–5 of the hierarchy, which contain relatively sparse numbers of codes, are analyzed separately to avoid dilution effects. An overview of the manufacturing sectors included in the survey is provided in Table 1, while Figure 2 summarizes the distribution of industry codes across sectors and levels.

2.1.3 Stationarity

Stationarity of indices is often regarded as a prerequisite for correlation analysis, since the presence of common trends can artificially inflate correlations by introducing dependence driven by trend alone.

In our dataset, however, not all industry codes are stationary. As illustrated in Figure 20, many industry codes show a stationary KLD index over time. Yet if we required full stationarity across all questions and time windows, no industry code would satisfy this condition.

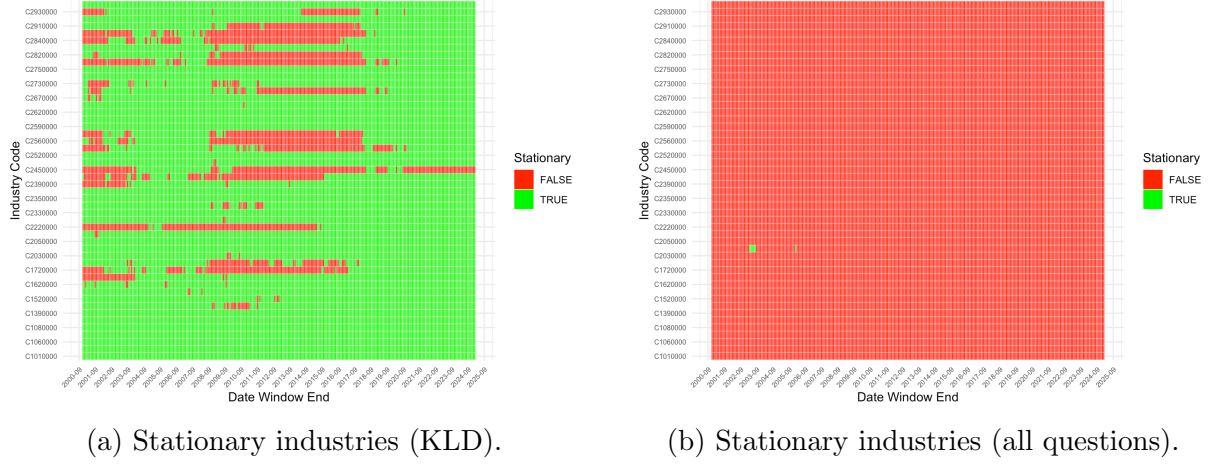


Figure 3: Comparison of stationarity across time windows and questions.

For the purposes of this study, we therefore did not impose stationarity as a strict requirement. While stationarity is generally desirable, excluding non-stationary series would eliminate too much information. Allowing for non-stationary indices may lead to higher measured correlations compared to the strict statistical definition, but in an economic context this is not necessarily problematic. In fact, when trends dominate certain rolling windows, correlations in trend may represent economically meaningful co-movement. From this perspective, we do not distinguish between “true” correlation and correlation driven by shared trends, as both can provide valuable insights into cyclical dynamics.

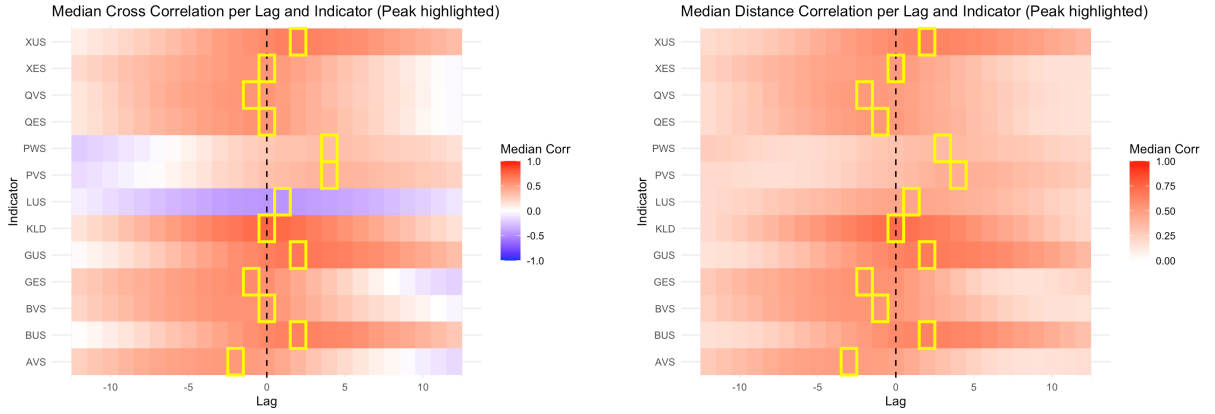
2.1.4 Correlation Metrics

To quantify the relationship between industry-level indicators and the main manufacturing index, we employ two complementary measures: *cross-correlation* and *distance correlation*. Using both metrics allows us to balance interpretability with sensitivity to nonlinear dependencies.

Cross-correlation: Cross-correlation, implemented via `stats::ccf`, measures the linear association between two time series at different leads and lags. It provides both the direction and magnitude of potential lead-lag structures. Because of its direct interpretability in terms of timing and sign, cross-correlation serves as the primary tool for detecting industries that anticipate or follow the dynamics of the main index.

Distance correlation: Distance correlation, implemented via `energy::dcor`, extends beyond linear relationships by detecting any form of statistical dependence, including nonlinear or nonmonotonic structures. A key property of distance correlation is that it equals zero if and only if the variables are independent. In our setting, it thus provides an upper bound relative to linear correlation ($|\rho|$) and signals whether cross-correlation may understate the strength of dependencies. The measure was introduced and studied in detail by ?.

Cross-correlation provides an intuitive baseline for interpreting the temporal structure of dependencies. Distance correlation, in turn, validates these findings by checking whether the relationships are predominantly linear or whether nonlinearities play a substantial role. In practice, both metrics yield strikingly similar structures (Figures 4a and 4b), suggesting that linear dependence accounts for the bulk of the observed relationships. Importantly, the combined figures also show that the strongest correlations are concentrated within a window of approximately ± 6 months. The peak correlation for each question consistently lies within this lag range. For this reason, all subsequent analyses focus on lags within ± 6 months, unless explicitly stated otherwise.¹



(a) Median cross-correlation profiles across industries.

(b) Median distance-correlation profiles across industries.

Figure 4: Comparison of median cross-correlation and distance-correlation profiles across industries.

2.2 Correlation Distribution

To gain a deeper understanding of the correlation structure between the survey questions and the main index, we analyze the distribution of correlations at the level of individual questions and industry codes. The goal of this exercise is to disentangle how strongly different industries contribute to the aggregate signal, and how stable these relationships are across time, sectors, and hierarchical levels.

Because the central point of interest here is the *strength of dependence*, we focus on distance correlation as the primary metric. Unlike cross-correlation, which emphasizes timing and direction, distance correlation allows us to summarize the overall magnitude of association independently of linearity or lag alignment. This makes it a natural choice for characterizing the distribution of correlations across a large number of industries.

2.2.1 Correlation Distribution by Time

From an economic perspective, it is important to assess whether the correlation structure remains stable across different phases of the business cycle, particularly during downturns

¹Rolling-window analyses introduced later confirm that this lag structure is stable across time and holds both in normal periods and during major economic disruptions.

and periods of stress compared to times of economic expansion. Figure 5 illustrates how average cross-correlations evolve across rolling windows, highlighting both the mean level and the location of peak lags.

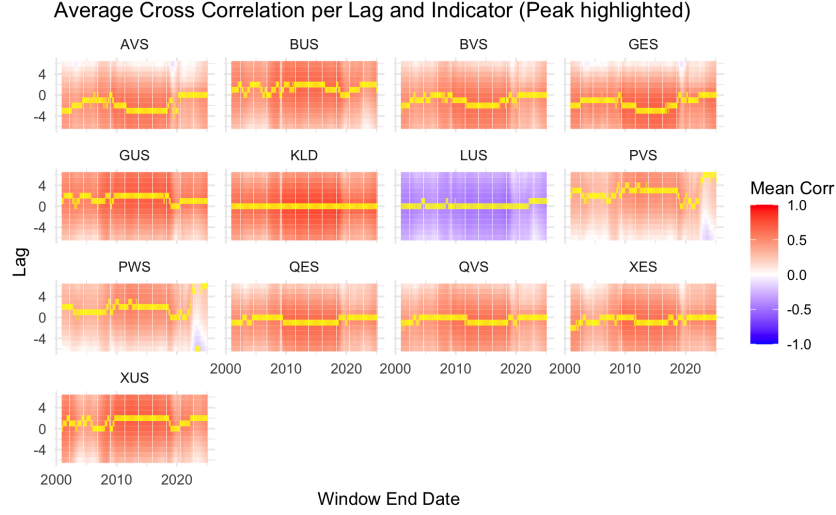


Figure 5: 10-year rolling window analysis (1-month step), 2000–2025.

Beyond average correlations, it is equally important to study the *distribution* of correlations across industries. Figure 6 presents the development of peak distance correlations over time, disaggregated by question. The figure shows both the median (red line) and the dispersion across industry codes (quantiles), thereby capturing not only the central tendency but also the variance of the dependence structure.

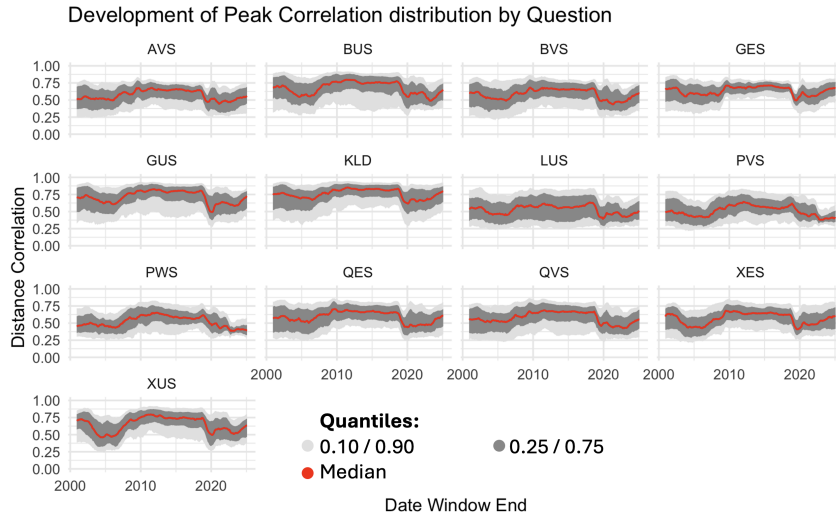


Figure 6: Peak distance correlation by question, with median and inter-quantile ranges, Levels 1–3.

Taken together, these results highlight two important insights. First, the correlation structure exhibits clear breaks around major economic events, most notably the global financial crisis of 2008–2009 and the COVID-19 pandemic after 2018. During these periods, both the level and the dispersion of correlations shift markedly, reflecting heterogeneous responses of industries to systemic shocks. Second, despite these disruptions, the overall correlation structure remains relatively stable in calmer periods, suggesting that the main index captures persistent common dynamics across manufacturing industries. This stability provides reassurance that industry-level correlations can serve as reliable indicators of aggregate developments, while the observed breaks emphasize their usefulness in detecting structural shifts.

2.2.2 Correlation Distribution by Sector

Examining the distribution of peak distance correlations at the sector level reveals substantial heterogeneity across industries (Figure 7). Some sectors exhibit consistently strong correlations with the main index, while others display weaker or more dispersed relationships. This underlines the heterogeneous character of the German manufacturing landscape, where not all industries contribute equally to the aggregate dynamics.

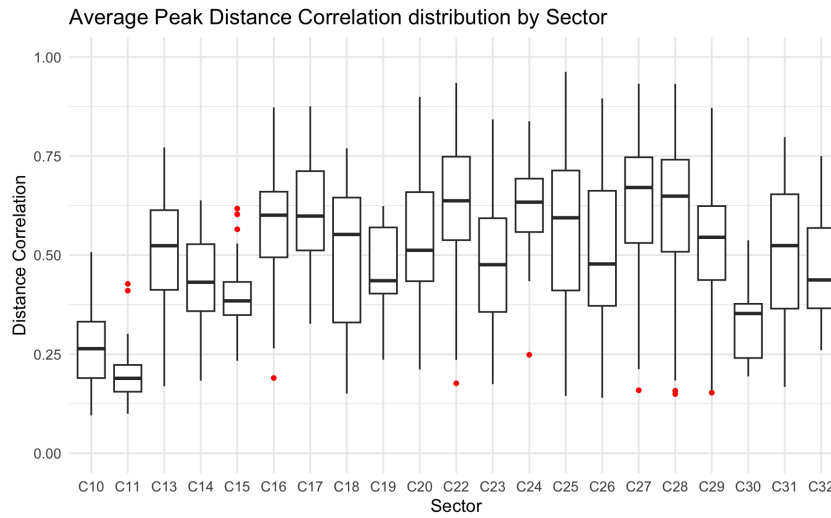


Figure 7: Distribution of peak distance correlations by sector.

It is important to note that the aggregation underlying the main index is weighted by the economic importance of each sector. As a result, the index successfully reflects overall economic developments, but it does not necessarily mirror the behavior of each individual industry. Certain sectors may therefore play only a limited role in shaping the aggregate signal, even though they are economically relevant in their own right.

From a modeling perspective, these results suggest that the dynamics of the main index can be effectively captured by focusing on a smaller set of industries with consistently strong correlations. Identifying such sectors provides a path towards sparser models that retain explanatory power while avoiding the noise introduced by less strongly aligned industries.

2.2.3 Correlation Distribution by Level

As discussed previously, the hierarchical structure of industry codes introduces an inherent imbalance in the number of series across levels. To account for this, we evaluate the correlation distribution separately for the more granular Levels 4–5. Figure 8 provides an overview of the distribution of peak distance correlations across levels.

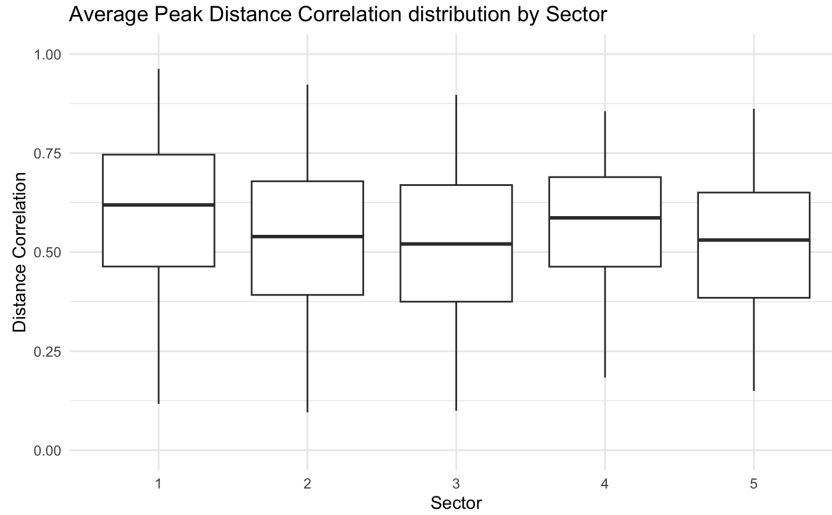


Figure 8: Distribution of peak distance correlations by hierarchical level.

The overall picture is that Levels 4–5 exhibit somewhat noisier correlation structures compared to the higher-level aggregations, but they broadly follow the same patterns observed at Levels 1–3. This similarity also extends to their temporal development (Figure 9), where the rolling-window evolution of distance correlations at deeper levels closely tracks that of the broader aggregates. In short, Levels 4–5 appear more volatile, but they remain aligned with the higher-level dynamics.

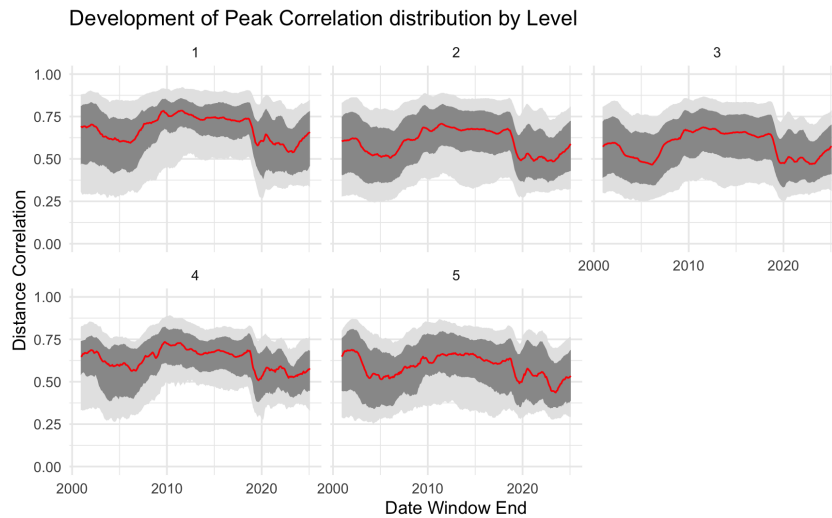


Figure 9: Rolling-window distribution of peak distance correlations by level.

From an analytical perspective, this is a noteworthy result. Because Levels 4–5 represent more disaggregated industries, their economic weight in the main index is much smaller than that of Level 1 or Level 2 sectors. Their alignment with the aggregate development is therefore not necessarily the result of construction, but likely reflects genuine causal links to overall business cycle dynamics. This suggests that lower-level industries may serve as a valuable indicative set for monitoring economic developments. Further investigation of these fine-grained series could thus provide useful insights into early signals of aggregate turning points.

2.3 Consensus Analysis

In the previous section we examined the variance of correlations across industries, which showed how strongly the dependence structure fluctuates over time. However, this analysis did not reveal whether the relative position of individual industries within the correlation structure remains stable or changes substantially. To address this, we introduce a ranking-based consensus analysis that evaluates the stability of industry rankings across time windows.

The idea is to compare, for each survey question and time window, the set of top- k most strongly correlated industries.² By focusing on the overlap of these top-ranked industries, we can track whether the “core set” of highly correlated industries remains consistent or whether entirely new industries enter the picture over time.

To quantify this, we compute the Jaccard index between two sets A and B of top- k industries:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{2k - |A \cap B|},^3$$

The Jaccard index ranges from 0 (no overlap) to 1 (perfect overlap), and thus measures how similar the top- k sets are across windows. Averaging these pairwise values over time yields a nonlinear consensus metric that captures the overall stability of the ranking structure.

²Throughout the analysis we use $k = 30$. This choice reflects a balance between capturing enough of the correlation structure and avoiding excessive noise from lower-ranked industries; robustness checks with alternative k confirmed that the dynamics are well represented at this threshold.

³The second equality holds only because both sets A and B are of equal size k . In the general case, the denominator is $|A \cup B|$ without simplification.

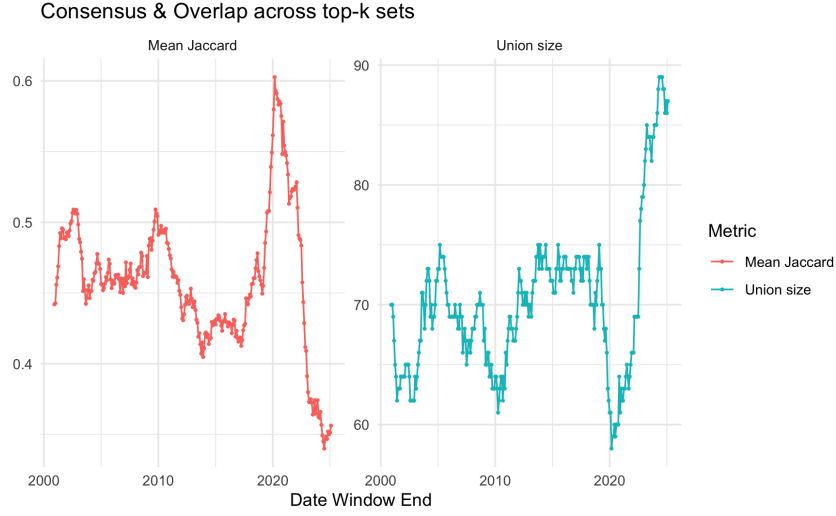


Figure 10: Consensus of top-30 sets (Jaccard index) vs. union size over time.

The results reveal two important features. First, higher average Jaccard values combined with smaller union sizes indicate that the structure is increasingly dominated by a relatively small group of industries, i.e., the same industries repeatedly occupy the top ranks. Second, we observe sharp spikes in consensus around major economic events. The strongest such spike occurs in 2020 during the COVID-19 pandemic, exceeding those seen during earlier crises such as the dot-com downturn in 2002 or the global financial crisis in 2008–2010. This pattern suggests that systemic shocks compress the correlation structure, temporarily aligning industries more closely than in normal times.

From an economic perspective, this alignment provides valuable information: during crises, the set of industries that best track the main index becomes more homogeneous, while in stable periods the ranking is more dispersed. In other words, systemic stress appears to “synchronize” industry-level dynamics. This observation points to the potential of consensus analysis as a diagnostic tool: sharp increases in consensus may serve as early signals of widespread economic stress. Further research should investigate whether such consensus spikes can be systematically linked to the onset of recessions or other structural breaks.

3 Granger Causality

3.1 Methodology

3.1.1 Motivation and Definition

This section moves beyond simple correlation to investigate linear predictive relationships between industry-specific time series and the main index. In particular, we are interested in finding industries that are "causing" the main index. Ideally, a cause should happen before its effect. Additionally, a cause should provide unique information about the future values of the effect. Finding such industries would enable us to forecast the main index. Similar considerations lead the economist Clive Granger to introduce the concept of "Granger Causality". Mathematically this is formalized by ? in the following way:

Definition 3.1. Let X, Y be time-series. X is said to (simple) Granger-cause Y iff.

$$\text{Var}(Y_{t+1}|Y_t, \dots, Y_1) > \text{Var}(Y_{t+1}|Y_t, \dots, Y_1, X_t, \dots, X_1)$$

We speak of instantaneous Granger Causality iff.

$$\text{Var}(Y_{t+1}|Y_t, \dots, Y_1) > \text{Var}(Y_{t+1}|Y_t, \dots, Y_1, X_{t+1}, \dots, X_1)$$

Note: Granger causality does not claim "true" or "philosophical" causality. Instead, it identifies predictive causality, a directional and temporal relationship in which one variable systematically precedes and improves forecasts of another.

3.1.2 Testing for Granger Causality

To test for Granger causality, the standard approach is to compare the predictive performance of two nested linear models. First, the dependent variable Y_t is regressed on its own past values up to a chosen lag order (auto-regressive model). Second, an extended model is estimated in which lagged values of another variable X_t are included in addition to the lagged values of Y_t . If the inclusion of X_t significantly improves the model's explanatory power, then X is said to Granger-cause Y .

Formally, the null hypothesis states that the lagged values of X do not improve the prediction of Y . The hypothesis is tested by comparing the restricted and unrestricted models using an F-test on the lagged X coefficients. If the null hypothesis is rejected at the chosen significance level, evidence for Granger causality is established. In practice, we use 5% as the significance level for the hypothesis test and a fixed number of lags (between 1 and 6). We use the standard *anova*-function from R to perform the hypothesis test.

3.2 Results

3.2.1 Instantaneous Granger Causality

We start our analysis with instantaneous Granger Causality as this concept has been longer established.

Test with single survey question

The analysis was performed on every single industry classified in level 2 and level 3. For each industry, we tested the relationship between individual survey questions and the main index, applying a range of time lags from one to six periods. This multi-lag approach ensures that we capture both immediate and delayed relationships between the variables.

- **Level 2:** For all 50 industries within this classification, we found that at least one survey question was instantaneous Granger Causal to the main index. This universal result at a broader industry level highlights a consistent predictive link.
- **Level 3:** The findings were similarly robust at the more granular level 3. Of the 61 industries analyzed, 60 demonstrated that at least one survey question was Granger Causal to the main index.

Identified Exception: A single exception was noted. The industry C2529000 - "Manufacture of containers" was the only one that did not show a Granger Causal relationship with the main index within the tested parameters.

Significant Survey Questions

Given that most industries show a predictive relationship with the main index, we next identify the specific survey questions that are the most consistent and powerful predictors.. By examining the distribution of instantaneous Granger Causality across all questions, we can determine which questions are most appropriate for use in forecasting models.

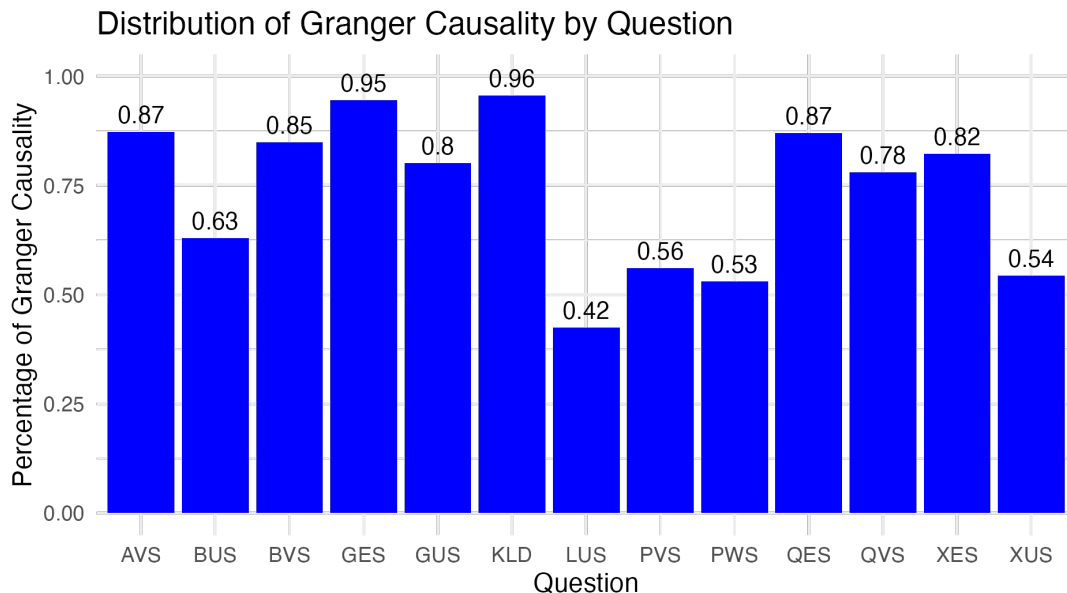


Figure 11: Distribution across questions indicating instantaneous Granger causality.

The key observations from the data are as follows:

- **Strongest Predictors:** A few questions stand out as exceptionally reliable. The question abbreviated as KLD shows an instantaneous Granger Causal relationship in 96% of the industries tested, followed closely by GES at 95%. Other strong performers include AVS and QES, both at 87%. These questions consistently provide predictive information across the vast majority of industries.
- **Weaker Predictors:** Not all questions carry the same predictive weight. The question LUS was found to be instantaneous Granger Causal in only 42% of industries, making it the least reliable predictor in this analysis. Other questions like PVS (56%) and PWS (53%) are also comparatively weaker, though still predictive for a majority of industries.

Magnitude of information gain

To quantify the marginal contribution of industry-specific data to the performance of the forecasting model, the adjusted R^2 was used as the evaluation metric. The adjusted R^2 is utilized to determine the proportion of variance in the aggregate index that is explained by the model, adjusting for the number of predictors. The "information gain" from incorporating industry-specific data was calculated as the delta in adjusted R^2 between a full model containing the industry data and a reduced baseline model without it. The analysis revealed a maximum increase in adjusted R^2 of only 0.0212, associated with the 'Manufacture of non-industry-specific machinery' sector and its business climate (KLD) question. This demonstrates that the marginal information gain from any single industry's data is negligible.

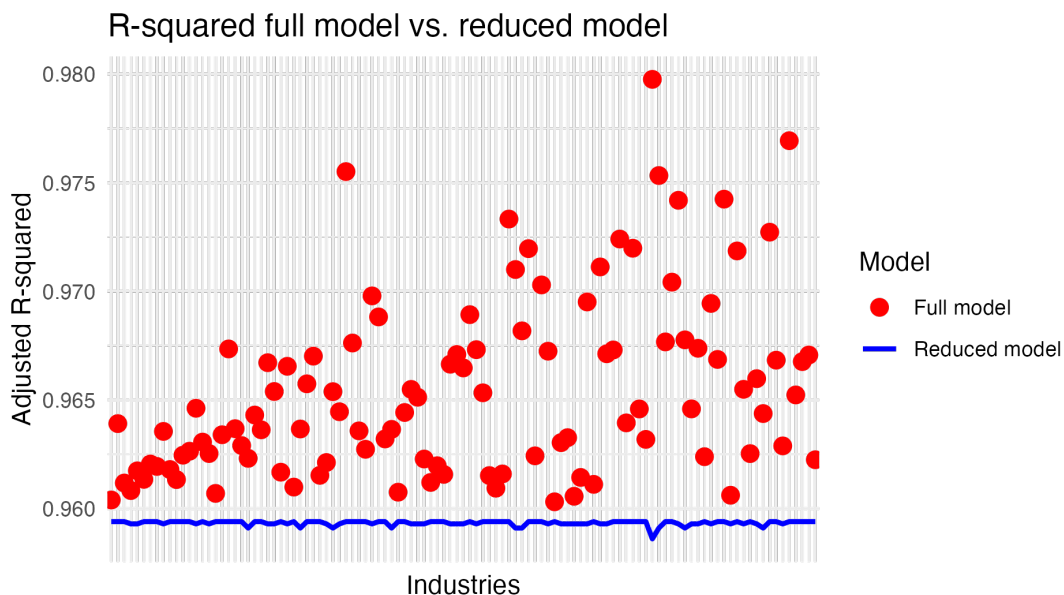


Figure 12: Adjusted R^2 differences (univariate).

Industry	Question	Lag	diff. adj. R^2
Manufacture of non-industry-specific machinery	Business climate	3	0.0212
Manufacture of parts and accessories for motor vehicles	Business climate	5	0.0175
Manufacture of plastic products	Business climate	5	0.0161
Manufacture of non-specialised machinery	Business climate	6	0.0149
Manufacture of machinery for other specific industries	Business climate	5	0.0148

Table 3: Top 5 industries by difference in adjusted R^2 (univariate).

Using all questions as covariates improves R^2 , but the same industries dominate.

Industry	Lag	diff. adj. R^2
Manufacture of non-industry-specific machinery	2	0.0247
Manufacture of plastic products	4	0.0224
Manufacture of parts and accessories for motor vehicles	6	0.0217
Manufacture of machinery for other specific industries	6	0.0204
Manufacture of non-specialised machinery	3	0.0191

Table 4: Top 5 industries by difference in adjusted R^2 (multivariate).

Visually, the business climate index for the 'Manufacture of Non-industry-specific machinery' sector exhibits a high correlation with the aggregate index but displays significantly greater amplitude in its cyclical fluctuations, with more pronounced peaks and troughs.

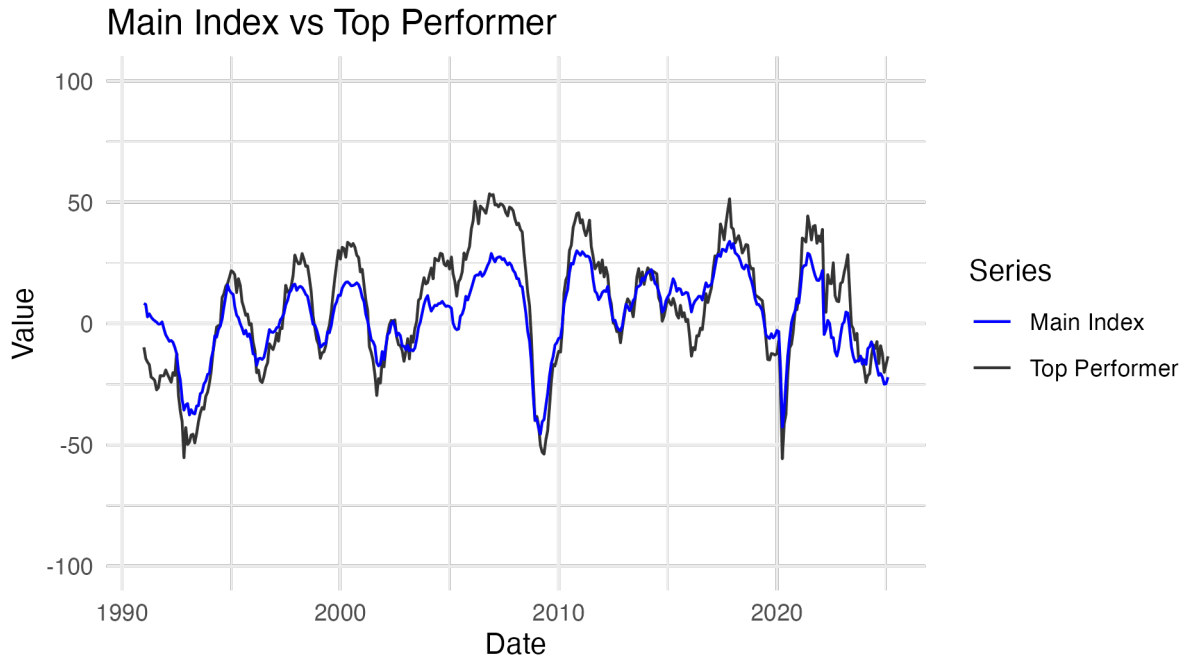


Figure 13: Main index vs. top-performing industry

3.2.2 Simple Granger Causality

Similar to the instantaneous case, the same analysis was also performed for simple Granger Causality on every single industry classified in level 2 and level 3. For each industry, we tested the relationship between individual survey questions and the main index, applying a range of time lags from one to six periods. For brevity we will highlight only the differences to instantaneous causality.

Test with single survey question

- **Level 2:** Only 48 out of 50 industries have at least one survey question that is Granger Causal to the main index.
- **Level 3:** Only 56 out of 61 industries have at least one survey question that is Granger Causal to the main index.

Significant Survey Questions

Not only do the industries that are Granger Causal change but also the percentage of how often a survey question is granger causal is almost halved. Also simple Granger Causality has the most information gain with a lag of 1.

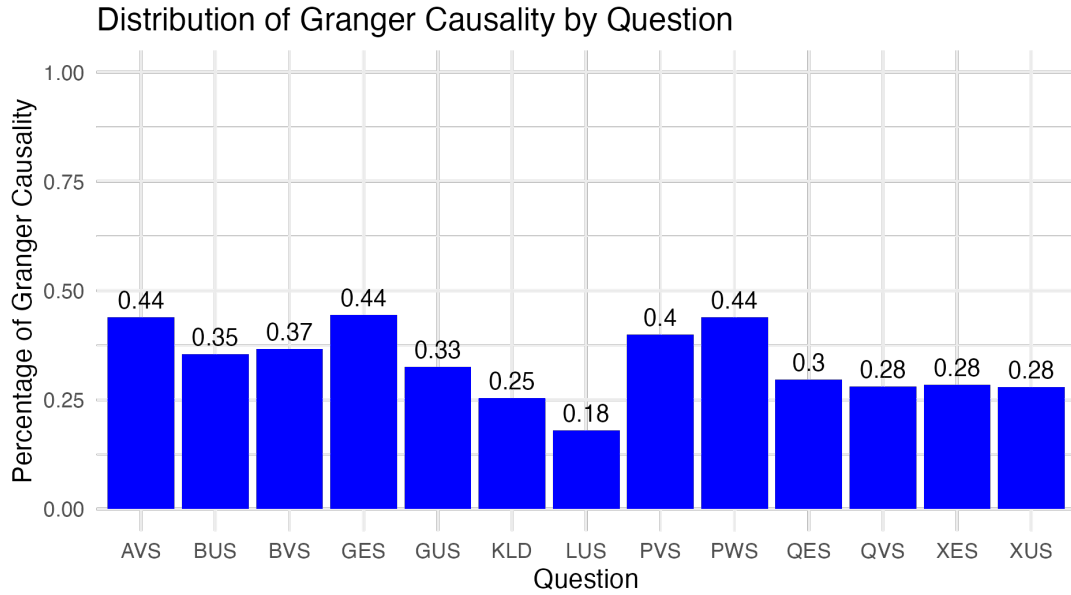


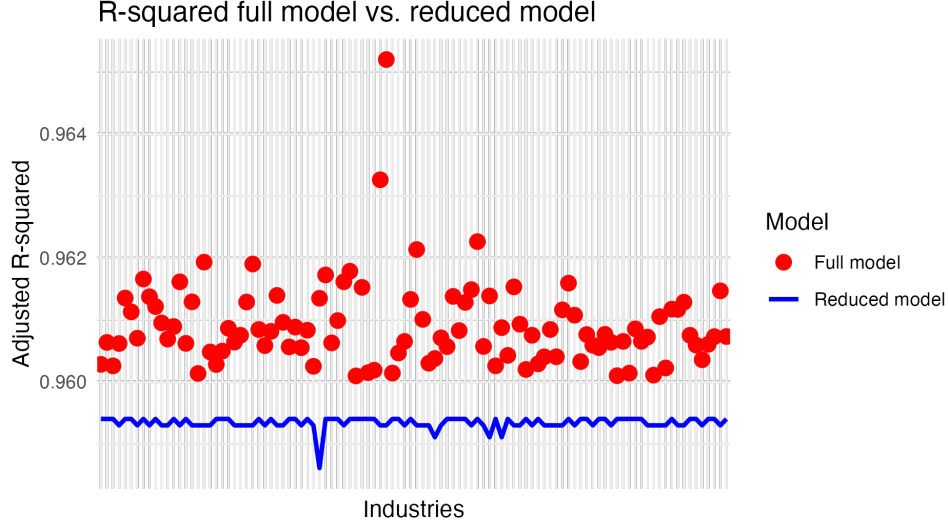
Figure 14: Distribution across questions indicating simple Granger causality.

The key observations from the data are as follows:

- **Strongest Predictors:** AVS, GES, and PWS at 44% are the strongest predictors with simple Granger Causality, but are far from the 96% maximum in instantaneous Granger Causality
- The predictive power of the KLD survey question for the main index is lost when relying solely on its past values. This suggests that while an industry-specific climate index may serve as a valid proxy for a contemporaneous assessment of the main index, it is insufficient for forecasting its future values.

Magnitude of information gain

Compared to instantaneous Granger Causality the maximum information gain in terms of adjusted R^2 is even lower at 0.0073 (compared to 0.0212)

Figure 15: Adjusted R^2 differences (univariate).

Industry	Question	Lag	diff. adj. R^2
Manufacturing of tools	business expectations	1	0.00730
Manufacturing of cutting devices, tools, and locks	business expectations	1	0.00714
Manufacturing of forged, pressed, drawn, and stamped parts	business expectations	1	0.00701
Manufacturing plastic goods (C2220000)	business expectations	1	0.00660
Manufacturing of other plastic goods (C2229000)	business expectations	1	0.00657

Table 5: Top 5 industries by difference in adjusted R^2 (univariate).

3.3 Conclusion for Granger Causality

Statistical analysis indicates that disaggregated survey data from a vast majority of individual industries, across both 2-digit and 3-digit classification levels, is Granger causal (simple and instantaneous) to the main-index. The robustness of this finding is underscored by the observation that this relationship was absent in only a few of the more than one hundred industries analyzed. However, while these relationships are statistically significant and do possess predictive power for the aggregate index, the marginal information gain from any single industry is negligible and therefore does not meaningfully improve the model's predictive performance. Additionally an implicit requirement for Granger causality is stationarity which is violated in some survey questions and industries as outlined in 2.1.3. Further analysis could examine groups of industries to predict the main index.

4 Indicative Subsets

We already established the existence of lead–lag structures between different questions and industry codes. A central goal is therefore to identify industries that allow us to predict the main index reliably. To enhance stability, we focus not on single industries but rather on subsets of industry codes.

The central idea is to identify subsets of industries that allow reliable prediction of the main index. To this end, we construct prediction sets based on lead–lag profiles. Specifically, we distinguish between a *Prediction Set*, which covers leads of one to six months, and an *Early Prediction Set*, which focuses on leads of four to six months. Industry codes are selected using different strategies, including sparse linear models, the scoring model, and clustering approaches. The resulting sets are then evaluated against benchmark models that rely solely on lagged values of the main index.

4.1 Elastic Net

The identification of smaller representative subsets can also be formulated as the problem of fitting a sparse regression model. A common way to induce sparsity is to include an ℓ_1 penalty term in the objective function. In this study we apply the Elastic Net, which combines an ℓ_1 penalty with an additional ℓ_2 penalty. The latter helps to stabilize the estimation process in the presence of multicollinearity, while the ℓ_1 component drives variable selection and promotes sparsity. The Elastic Net objective is given by

$$\min_{\beta_0, \beta} \frac{1}{2N} \sum_{i=1}^n (y_i - \beta_0 - x_i^\top \beta)^2 + \lambda \left((1 - \alpha) \|\beta\|_2^2 + \alpha \|\beta\|_1 \right),$$

where λ controls the overall degree of regularization and α determines the balance between the ℓ_1 and ℓ_2 components.

Because the number of observations is relatively small compared to the large set of potential predictors, feature preselection was applied to improve stability and interpretability of the results. In particular, we restricted the predictor set to industry codes at level 2, which provides a manageable yet sufficiently rich representation of sectoral dynamics.

4.2 Scoring Model

Two challenges we faced when selecting industry codes for predictions is that Simple correlations between industry codes and the main index vary substantially across windows and questions, and rank-based approaches often inflate the size of selected sets when consensus is low. To address this, we develop a scoring model that provides a stable and comparable measure of predictive power across time and survey questions. The goal is to assign each industry code a single score that reflects its predictive relevance and allows for consistent selection of the top- k industries.

The scoring model is designed to capture both the strength and the variability of correlations, while also reflecting structural patterns across questions and windows. Importantly, it provides not only a ranking of industries but also meaningful distances between them.

For each window t , question q , and industry i , we compute the peak correlation $c_{i,t,q}$. We then define the relative performance

$$r_{i,t,q} = c_{i,t,q} - \max_j c_{j,t,q},$$

and aggregate across all windows and questions to obtain

$$\text{Score}_i = \frac{1}{TQ} \sum_{t,q} r_{i,t,q}^2.$$

This formulation yields stable scores, where the squared deviations measure the distance of each industry to the pointwise best performer.

As illustrated in Figure 16, the average and minimum scores for the top-ranked industries are very close to one another, while towards the lower end the scores drop steeply. This pattern indicates that although predictive performance is relatively strong across a wide range of industries, a smaller subset consistently stands out. The scoring model thus provides a principled way to identify predictive sets and complements the other selection approaches.

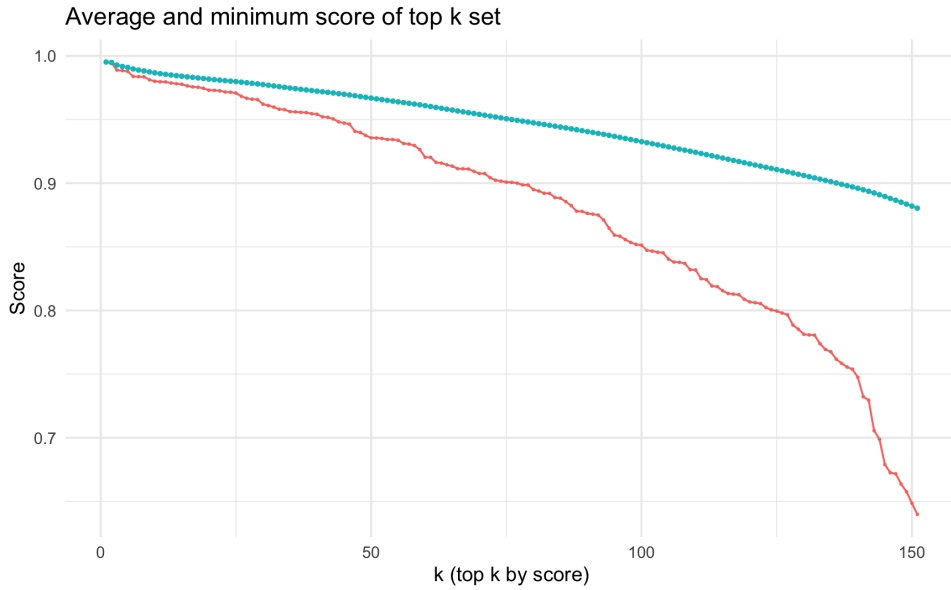


Figure 16: Prediction performance (Prediction Set).

4.3 Model Performance

In the following, we compare the performance of the industry codes selected by the Elastic Net and the Scoring Model. For both approaches, we fit a linear model that includes the selected industry codes together with the lagged main index. Performance is evaluated in terms of mean squared error (MSE), root mean squared error (RMSE), and the coefficient of determination (R^2).

4.3.1 Prediction Set

For the scoring model we selected the five highest-scoring industry codes at level 2. The selected industries are listed in Table 6.

Table 6: Industry codes selected by the Scoring Model

Code	Name
C1620000	Manufacture of other products of wood; manufacture of articles of cork, straw and plaiting materials
C1720000	Manufacture of articles of paper and paperboard
C2220000	Manufacture of plastic products
C2440000	Manufacture of basic precious and other non-ferrous metals
C2890000	Manufacture of other special-purpose machinery

These industries are predominantly associated with packaging materials, intermediate goods, and investment goods such as machinery. From an economic perspective, this selection is plausible: packaging and input materials are required early in the production process, and machinery is ordered in anticipation of future demand. Consequently, these sectors are likely to lead the overall economic development, with orders placed in advance during expansions and cancelled early in recessions.

The Elastic Net procedure selected a somewhat different set of industries, shown in Table 7. While the codes differ from those identified by the scoring model, the economic interpretation is similar: these industries also represent upstream or investment-oriented activities that tend to react early to changes in the business cycle.

Table 7: Industry codes selected by the Elastic Net

Code	Name
C2550000	Forging, pressing, stamping and roll-forming of metal; powder metal-lurgy
C2560000	Treatment and coating of metals; machining
C2840000	Manufacture of machine tools
C2920000	Manufacture of bodies (coachwork) for motor vehicles; manufacture of trailers and semitrailers

Because the models differ in the number of predictors, we evaluate performance using R^2 rather than adjusted R^2 . The results are reported in Table 8. Both selection methods achieve a higher fit than the benchmark model based solely on the lagged main index. The scoring model performs slightly better than the Elastic Net, although the differences are modest. This outcome is in line with expectations: given its smaller set of selected

features, the Elastic Net is likely to perform somewhat worse than the scoring model but still slightly better than the lagged benchmark.

Table 8: Prediction performance (Prediction Set)

	MSE	RMSE	R^2
Scoring Model	9.48	3.08	0.97
Elastic Net	9.99	3.16	0.96
Lagged Main Index	11.00	3.32	0.96

Figure 17 plots the model predictions against the observed main index. As expected, all models track the actual development closely. This is not surprising, since a lead of one month is economically short and provides relatively limited scope for divergence.

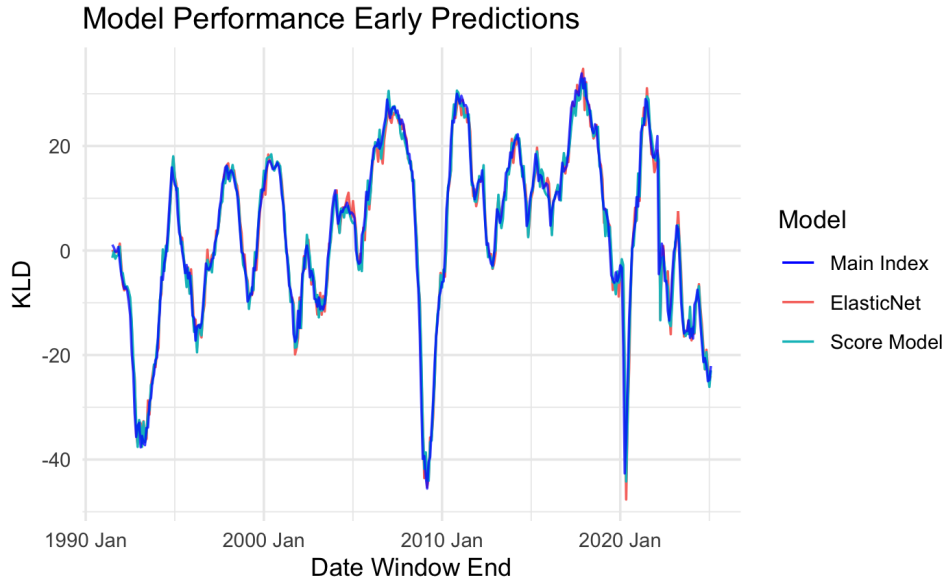


Figure 17: Prediction performance of different models (Prediction Set).

4.3.2 Early Prediction Set

For the early prediction set we restricted the lead to four to six months, thereby focusing on industries that provide signals further in advance. The selected industries are shown in Tables 9 and 10. While the sets partly differ from those in the general prediction set, there is notable overlap in the types of sectors identified, underlining the same economic reasoning: industries producing upstream inputs or investment goods tend to react early to changes in demand expectations.

Table 9: Industry codes selected by the Scoring Model (Early Prediction Set)

Code	Name
C1610000	Sawmilling and planing of wood
C1710000	Manufacture of pulp, paper and paperboard
C1720000	Manufacture of articles of paper and paperboard
C2220000	Manufacture of plastic products
C2440000	Manufacture of basic precious and other non-ferrous metals

Table 10: Industry codes selected by the Elastic Net (Early Prediction Set)

Code	Name
C1610000	Sawmilling and planing of wood
C2550000	Forging, pressing, stamping and roll-forming of metal; powder metal-lurgy
C2560000	Treatment and coating of metals; machining
C2920000	Manufacture of bodies (coachwork) for motor vehicles; manufacture of trailers and semitrailers

As in the general prediction set, we evaluate the performance of the scoring model, the Elastic Net, and the benchmark model based on lagged values of the main index. The results are summarized in Table 11. Both selection methods outperform the benchmark, with the scoring model again achieving slightly better predictive accuracy than the Elastic Net.

Table 11: Prediction performance (Early Prediction Set)

	MSE	RMSE	R^2
Scoring Model	63.00	7.94	0.77
Elastic Net	66.00	8.12	0.76
Lagged Main Index	79.40	8.91	0.71

Figure 18 plots the model predictions against the observed main index. Compared to the one-month lead predictions, the increased horizon naturally leads to a lower overall fit. Nonetheless, both the scoring model and the Elastic Net improve substantially over the lagged benchmark, confirming the usefulness of early-predictive industry codes.

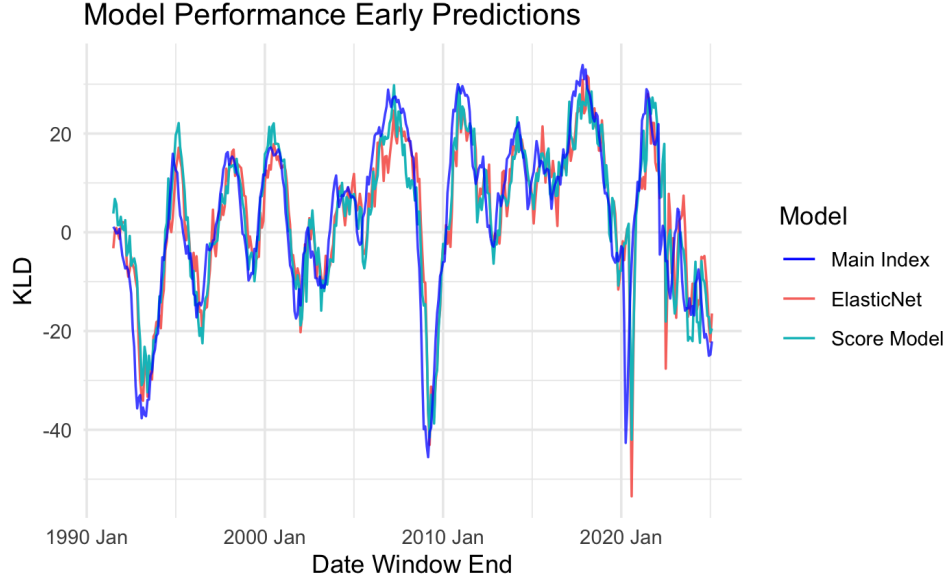


Figure 18: Prediction performance of different models (Early Prediction Set).

4.3.3 Performance Comparison and Interpretation

The comparison of cross-correlation profiles highlights important differences between the two selection approaches. As shown in Figures 19 and 20, the Sparse LM (Elastic Net) identifies industry codes with stronger lead properties, but at the cost of weaker overall correlations. By contrast, the scoring model yields a more robust correlation structure, although peak correlations tend to occur at shorter leads. This reflects the design of the scoring model: by penalizing outliers more heavily, it favors industry codes that exhibit consistently strong correlations across questions and windows, thereby providing a more “general” solution.

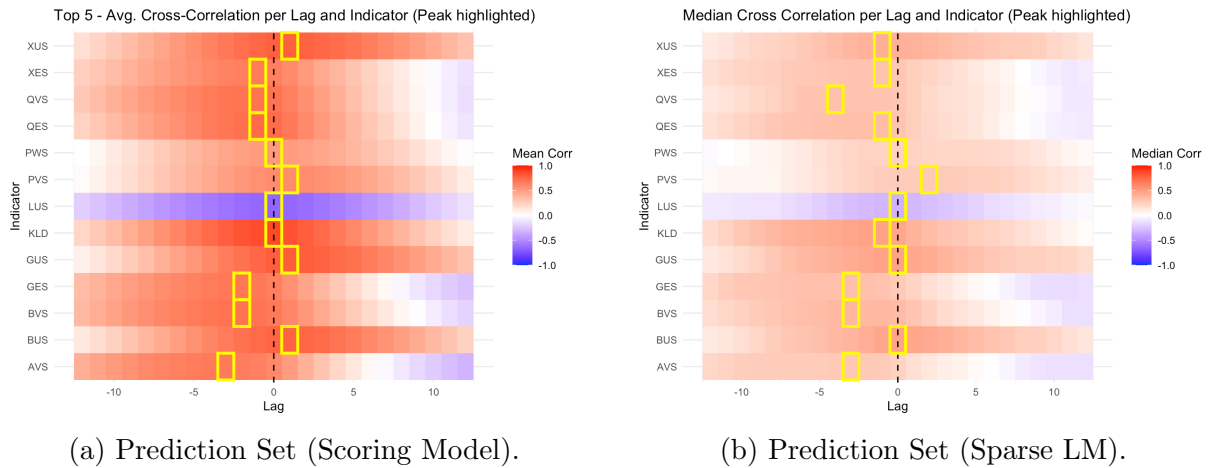


Figure 19: Comparison of median cross-correlation profiles across industries for the Prediction Set. Yellow boxes indicate peak lags; vertical dashed line marks zero.

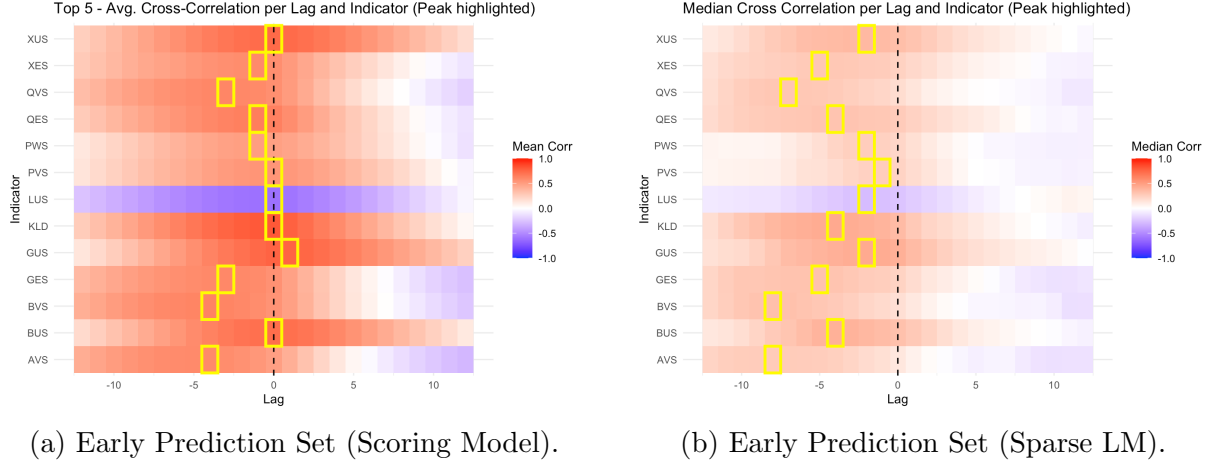


Figure 20: Comparison of median cross-correlation profiles across industries for the Early Prediction Set (lead 4–6 months).

Lagging questions such as BUS and GUS show relatively low lags of peak correlation, illustrating how the two strategies emphasize distinct aspects of predictive performance. The scoring model captures industries with dynamics broadly similar to the main index, which results in highly correlated predictor sets. The ℓ_1 component underlying the Sparse LM encourages selection of more distinct predictors by retaining one representative from groups of highly correlated industries and discarding the rest; a small ℓ_2 component (Elastic Net) is included to stabilize estimation under multicollinearity (cf. ?). This methodological difference helps explain why the selected subsets diverge and why the average correlation strength varies between methods.

To assess robustness, we additionally compare both methods to 500 random subsets of five level-2 industry codes. Figures 21a and 21b display the distribution of R^2 across these random subsets. As expected, nearly all subsets outperform the benchmark based solely on lagged main-index values. The scoring model consistently ranks among the stronger subsets, supporting its usefulness as a versatile selection method for predictive modeling.

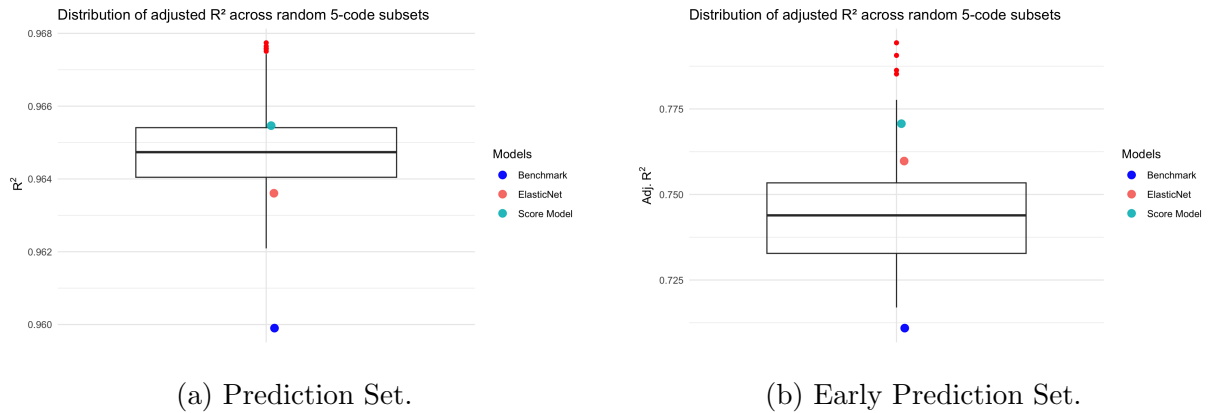


Figure 21: Distribution of R^2 across 500 random 5-code subsets, with benchmark (lagged main index), Elastic Net, and Scoring Model overlaid.

At the top end of the distribution, performance is dominated by a small number of outlier subsets. Tables 12 and 13 list the three best-performing random subsets for the Prediction Set and the Early Prediction Set, respectively. Notably, each table contains twelve distinct industry codes across only three subsets (3×5), indicating that strong predictive performance is not confined to a unique combination of industries. Rather, what matters are shared economic characteristics: upstream inputs, packaging/intermediate goods, and investment-related sectors tend to provide early signals of cyclical turning points.

Table 12: Top 3 random subsets (Prediction Set)

Industry Codes	R^2	RMSE
C1720000, C2210000, C2590000, C2450000, C2390000	0.968	2.98
C1610000, C2740000, C3250000, C2710000, C2450000	0.968	2.98
C2570000, C2440000, C1720000, C2310000, C2710000	0.968	2.98

Table 13: Top 3 random subsets (Early Prediction Set)

Industry Codes	R^2	RMSE
C1610000, C2920000, C2910000, C2930000, C1060000	0.794	7.52
C2610000, C1610000, C2340000, C1710000, C2310000	0.791	7.58
C2550000, C2340000, C3250000, C1610000, C2040000	0.786	7.66

Multiple paths exist to construct effective predictive sets. The scoring model offers stability and interpretability by selecting industries with consistently strong correlations to the main index across questions and windows. The Elastic Net (Sparse LM) promotes diversity by emphasizing distinct predictors and stabilizes estimation via an additional ℓ_2 component. Random subsets demonstrate that predictive skill is widely distributed across industries, yet especially pronounced among upstream and investment-oriented sectors.

4.4 Clustering based on CCF

So far, we have mostly considered all questions within each industry. To find potentially leading groups, which can contain different questions of different industries, we took on a clustering approach based their relationship with the main index. We computed the CCF of all individual questions of all level 2 aggregates and then performed Ward’s method to cluster them (?). In a nutshell, Ward’s method starts with each component as its own cluster and iteratively merges clusters in a way that minimizes the overall within-cluster variance.

4.4.1 Comparing cluster groups

A challenge lied in finding a good measure to compare the resulting groups and rank out potentially interesting ones. Simply ranking clusters by correlation strength does not distinguish between leading and lagging behavior. To address this, we adopted a similar strategy as in the previous methods: we fitted a linear model predicting the main index y using its own lagged values along with the lagged values of a cluster average x up to lag k :

$$y_t \sim y_{t-1} + y_{t-2} + \dots + y_{t-k} + x_{t-1} + x_{t-2} + \dots + x_{t-k}$$

Again, we used one setting with all lagged values up to 6 months behind and another early setting, where we only considered lags 4 to 6. Table 14 shows the top 5 clusters according to the MSE of the above model for both settings.

Cluster	n	MSE	RMSE	Adj. R^2	Cluster	n	MSE	RMSE	Adj. R^2
2	9	10.27	3.21	0.9615	63	8	65.39	8.09	0.758
10	9	10.31	3.21	0.9613	48	4	66.18	8.14	0.755
16	7	10.32	3.21	0.9613	1	8	67.46	8.21	0.751
63	8	10.37	3.22	0.9611	3	12	67.72	8.23	0.750
62	5	10.41	3.23	0.9609	5	10	69.97	8.36	0.741
Lagged Main	1	11.00	3.32	0.959	Lagged Main	1	79.4	8.91	0.70

(a) Considering lags 1 to 6.
(b) Considering lags 4 to 6

Table 14: Top 5 clusters based on MSE

The predictive performance of the above model alone is not necessarily a sufficient indicator to find a meaningfully leading group. Lagged values of a cluster average could add relevant information to the lagged values of the main index, resulting in a low MSE, while still not leading the main index. A visual examination of the CCFs of the cluster can help in further determining leading clusters. Fig. 22 shows the CCFs of the clusters from table 14. As expected, more of the top 5 clusters of the early prediction setting seem to be leading, compared to the other setting.

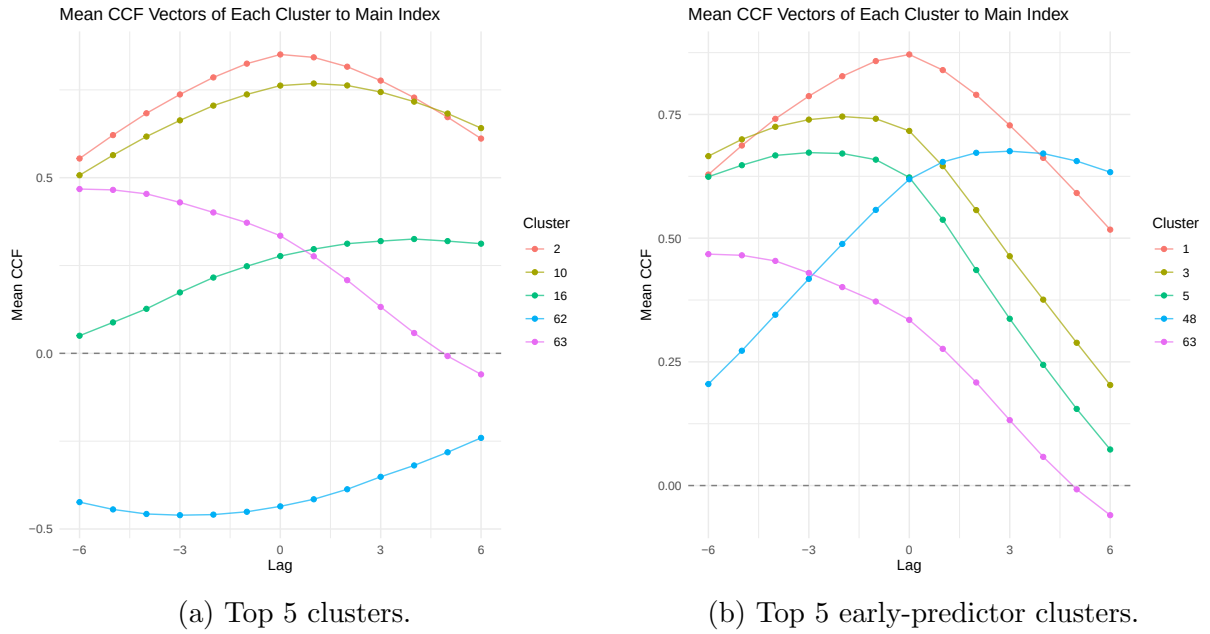


Figure 22: Mean CCFs of the top 5 cluster averages.

4.4.2 Exemplary Cluster

Although most clusters are difficult to interpret, cluster 3 from the top 5 provides an illustrative example of a potentially meaningful grouping. As shown in Fig. 23, the series in this cluster tend to lead the main index, while Table 15 details its constituent components. Its dominated by questions which compare the current situation to the previous month and many of the industries deal with intermediate components or capital goods.

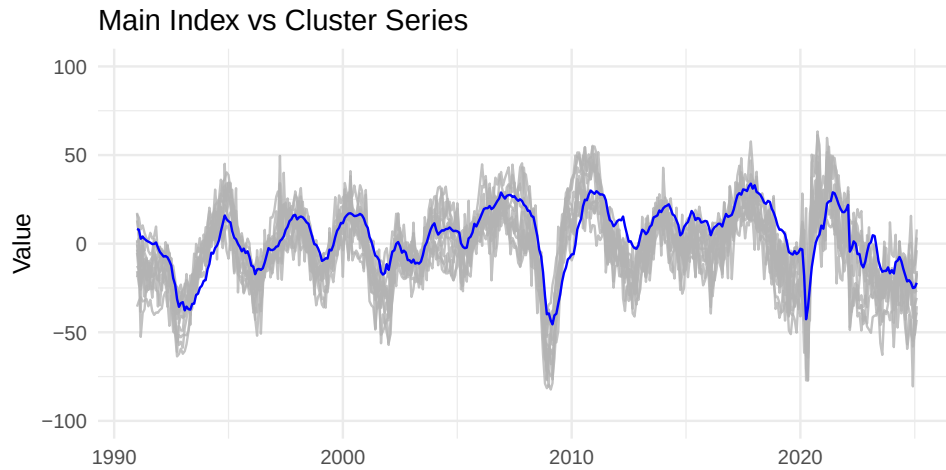


Figure 23: Main index (blue) vs individual series in cluster 3.

Industry Title	Question Title
Manufacture of plastic products	Business situation expectations
Manufacture of plastic products	Order backlog compared to previous month
Manufacture of plastic products	Production compared to previous month
Manufacture of forged, pressed, drawn, and stamped parts	Order backlog compared to previous month
Manufacture of forged, pressed, drawn, and stamped parts	Production compared to previous month
Manufacture of forged, pressed, drawn, and stamped parts	Production plans
Manufacture of cutlery, tools, and locks	Business situation expectations
Manufacture of cutlery, tools, and locks	Demand compared to previous month
Manufacture of electric motors, generators, transformers, etc.	Demand compared to previous month
Manufacture of non-industry-specific machinery	Demand compared to previous month
Manufacture of machine tools	Business situation expectations
Paper, cardboard, and paperboard processing	Order backlog compared to previous month

Table 15: Individual series present in cluster 3.

5 Turning Point Indication

In this final section, we are gonna take a look at whether we can indicate not the main index itself, but rather its turning points. There are several different definitions of economic turning points, and no single standard exists for Germany. We apply the Bry–Boschan algorithm (?), as implemented in the R package BCDating (?), to identify the troughs and peaks of the main index, which serve as the groundtruth. The algorithm analyzes the data retrospectively and identifies turning points based on several criteria, such as minimum cycle or phase lengths. A trough corresponds to an expansion turning point, indicating that the economy is about to enter a growth phase, commonly referred to as a boom. Conversely, a peak signals the end of the previous expansion and the beginning of a contraction, typically known as a recession.

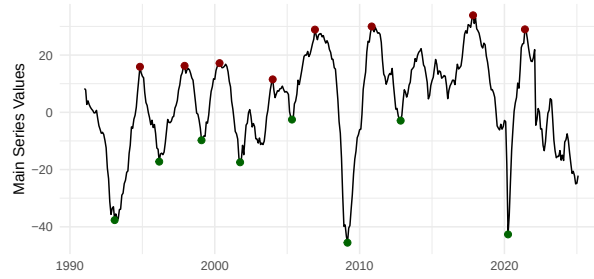


Figure 24: Bry-Boschan Turning Points of the main index.

5.1 Markov-Regime-Switching

To simulate a realistic real-time setting for potentially identifying subseries, the Bry-Boschan algorithm is not suitable, as it requires the full data range. Instead, we apply a two-regime Markov-switching model (?), which allows the system to switch between two regimes, s_t , each month and follow the implementation of the R package MSwM (?) for the model. The probability of switching is governed by a latent Markov chain, and each regime is characterized by a different intercept:

$$y_t = \mu_{s_t} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2),$$

Each month, the model provides the probability of switching to the upper regime with a higher intercept in the following month. Following ?, we apply a decision rule to classify turning points: if the probability of the upper regime exceeds 0.66 for the first time, it is considered an expansion turning point and if it falls below 0.33 for the first time, it indicates a contraction turning point. Probabilities between 0.33 and 0.66 are treated as uncertain and do not form a strong enough signal for a turning point yet.

The above model can either be run with smoothed probabilities, meaning that the whole data range is used to fit the model, or filtered probabilities, where only data up to the current month t is used. The later one simulates a real-time forecasting scenario and is thus suitable for our evaluation. Note that we did not perform any train-test split to evaluate forecasting models, this is rather a descriptive setting based on the past 30 years.

5.2 Classification Task

Fitting the above to all individual level 1 series gives us turning points for all of these series. To evaluate whether these individual turning points, predicted in real-time, can indicate the groundtruth turning points of the main index, we see it as a classification task. A hit occurs if an actual main turning point follows within a fixed time window after a predicted one. Table 16a shows the typical classification metrics we used. However, it also shows how no individual series did bring up any meaningful results, even when tolerating a three months delay for the predicted turning points, compared to the actual ones.

Industry	Question	Hits	Total Pred	Total Actual	Precision	Recall	F1	Lead Time (sd)
C1900000	PVS	7	47	16	0.151	0.438	0.224	2.96 (2.76)
C2000000	LUS	4	20	16	0.202	0.250	0.223	5.00 (1.41)
C1300000	PWS	4	20	16	0.200	0.250	0.222	4.00 (2.00)

(a) 0 - 6 Months Lead time

Industry	Question	Hits	Total Pred	Total Actual	Precision	Recall	F1	Lead Time (sd)
C2000000	QES	5	21	16	0.312	0.312	0.312	-0.60 (2.88)
C1900000	PVS	9	47	16	0.196	0.562	0.291	2.17 (3.05)
C1900000	PWS	6	27	16	0.225	0.375	0.280	2.00 (3.34)

(b) -3 - 6 Months Lead time

Since individual series did not appear to provide meaningful signals, we explored grouping multiple series to see if a combined signal could yield a reliable indication. We tested several approaches for aggregating turning points across series: collecting turning points within a rolling window and signaling if a certain threshold was exceeded; averaging their binary states based on the most recent turning point; and averaging the Markov-switching probabilities across series, followed by applying the 0.33/0.66 decision rule.

We also used several different approaches to form the groups, such as simply aligning series by their lag value with peak correlation or switching to level 2 and using the groups obtained from CCF-based clustering. However, none of these approaches produced meaningful or interpretable signals.

5.2.1 Grouped Series - Flipped

However, a small effect was observed when noting that, for some groups, the approximate number of turning points roughly matched the number of ground-truth turning points. For instance, considering all series with peak CCFs at lags -1, 0, and 1, so series with roughly synchronous characteristics. Their turning points were lagging quite far behind the Bry-Boschan turning points of the main index, potentially also simply due to the nature of using a Markov switching model with differing intercepts. But flipping the prediction goal and predicting opposite turning points, predicting contraction points as expansions and vice versa, the subseries do seem to roughly indicate main turning points. Whether this has any actual value is open, but a potential interpretation might be that once even these

groups have reached the current regime according to their Markov switching probability, it might soon be over and a turning point could occur in the next twelve months. Fig. 25 shows the turning points of the above group compared to the main ones and table 17 the scores when evaluating according to this flipped scenario.

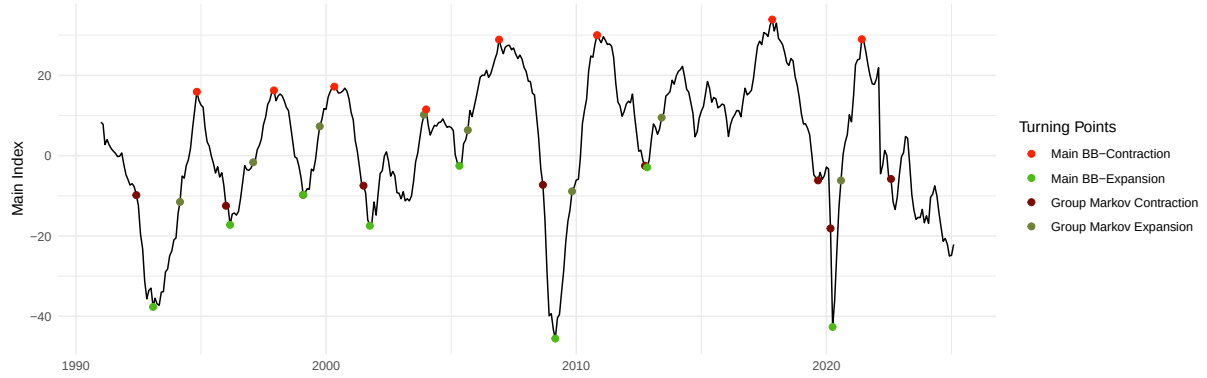


Figure 25: Turning points of group of all level 1 series with roughly composite moving characteristics according to their peak CCF lag value.

Main Type	Hits	Total Pred	Total Actual	Precision	Recall	F1	Lead Time (sd)
Expansion	6	7	8	0.857	0.75	0.8	5.67 (3.78)
Contraction	6	8	8	0.75	0.75	0.75	3.17 (3.37)

Table 17: Evaluation of the flipped indication setting.

6 Conclusion

This consulting project examined the internal structures of the ifo Business Survey (1991–2025) with the aim of detecting systematic relationships between the main index and the underlying time series. Rather than building a full forecasting model, the focus was on identifying small *predictive subsets* of industries which stick out as especially important for the overall index. We find strong correlations with clear lead–lag structures within ± 6 months that compress during crises, pointing to their role as early stress signals. While most industries are Granger causal, their individual contribution is negligible, which emphasizes the importance of subsets and clusters. The time series showing up across the various grouping and clustering approaches of our analysis often contained industries which are rather early in the supply chain or investment related.

Correlation Analysis The analysis reveals strong dependencies between industry indicators and the main manufacturing index, concentrated within a ± 6 month window. Despite sectoral imbalance and non-stationarity, the correlation structure remains coherent and economically meaningful. Consensus results further show that correlations compress during crises, suggesting that shifts in industry alignment may provide early signals of systemic stress and warrant deeper investigation.

Granger Causality Almost all manufacturing industries are Granger causal (simple and instantaneous) to the aggregate main-index. The contribution of any single industry is minimal in terms of information gain measured by adjusted R^2 . Additionally an implicit requirement for Granger causality is stationarity which is violated in some survey questions and industries (see 2.1.3). Thus, we find this method to be rather unsuitable for further analysis and insights.

Elastic Net vs. Scoring Model. Both methods improve on the benchmark of lagged main-index values. Compared to 500 random five-industry subsets, they perform better in the early prediction set, likely because overfitting is harder there, while the general prediction set leaves more scope for spurious fit. Although the exact industry codes differ, both approaches consistently select sectors linked to inputs, packaging, and investment goods. From an economic perspective, these sectors are upstream in the value chain and therefore tend to adjust production earlier, making them natural candidates for anticipating aggregate manufacturing dynamics.

CCF clustering. We also clustered individual survey questions (level 2) by their cross-correlation with the main index. To compare the clusters, we fitted linear models with lagged values of cluster average cluster and lagged values of the main index itself to the main index. We combined it with visually assessing the cross-correlation function to identify potentially leading and interesting clusters. Especially the early prediction setting brought out some leading clusters. While many clusters are hard to interpret, clearer examples feature intermediate goods and investment-related activities.

Turning points. Trying to explicitly indicate turning points and not just the main index itself was evaluated separately. We used the Bry-Boschan algorithm to identify turning points of the main index and Markov Switching models for potentially indicating turning points of the subseries. Neither individual nor grouped series were able to reliably indicate the Bry-Boschan turning points of the main index, when modeling their levels with Markov Switching. Further smoothing the series and applying Markov switching models to model the first differences or other more sophisticated approaches could be examined in further analysis.

Outlook. Taken together, the results show that the underlying time series of the ifo Business climate are able to provide meaningful information beyond the main index alone. Future work could explore the data in greater depth, as many of our results suggested points to further elaborate on. Also, our analysis was restricted to the monthly questions of the manufacturing industry. Taking all of the 40.000 time series of the survey into account, could lead to many more insights.

7 Contributions

Jakob Freytag wrote the Introduction, the Turning Point Indication chapters, the CCF based clustering and the respective code. He also coded the data pre-processing. Dominik Pitz wrote the chapters on Correlation Analysis and Indicative Subsets with the respective Code. Daniel Gloukhman wrote the chapter on Granger Causality and the respective code. He also wrote the code to import and combine the ifo-dataset.

We used generative AI-tools like ChatGPT and Gemini for refining grammar, text style and improving readability of this report. Further, these tools also aided us in coding and especially in commenting the code.

8 Appendix

Table 18: Industry Codes and Titles

Industry Code	German Title	English Title
C0000000	Verarbeitendes Gewerbe	Manufacturing
CONG0000	Verarbeitendes Gewerbe (ohne Ernährungsgewerbe)	Manufacturing (excluding food industry)
C00000A0	Herstellung von Vorleistungsgütern	Manufacture of intermediate goods
C00000B0	Herstellung von Investitionsgütern	Manufacture of capital goods
C00000C0	Herstellung von Konsumgütern (Ge- und Verbrauchsgüter)	Manufacture of consumer goods (durable and non-durable)
C0000CD0	Herstellung von Gebrauchsgütern	Manufacture of durable goods
C0000CN0	Herstellung von Verbrauchsgütern	Manufacture of non-durable goods
CONG00C0	Herstellung von Konsumgütern (ohne Ernährungsgewerbe)	Manufacture of consumer goods (excluding food industry)
C2220000	Herstellung von Kunststoffwaren	Manufacture of plastic products
C2221000	Herstellung von Platten Folien Schläuchen und Profilen	Manufacture of plates, sheets, tubes and profiles
C2222000	Herstellung von Verpackungsmitteln aus Kunststoffen	Manufacture of plastic packaging
C2223000	Herstellung von Baubedarfsartikeln aus Kunststoffen	Manufacture of plastic building products
C2229000	Herstellung von sonstigen Kunststoffwaren	Manufacture of other plastic products
C2300000	Glas- Keramikgewerbe Verarbeitung von Steinen und Erden	Manufacture of glass, ceramics, processing of stones and earths
C2310000	Herstellung von Glas und Glaswaren	Manufacture of glass and glass products
C2312000	Veredelung und Bearbeitung von Flachglas	Finishing and processing of flat glass
C2313000	Herstellung von Hohlglas	Manufacture of hollow glass
C2319000	Herstellung von technischen Glaswaren	Manufacture of technical glassware

Continued on next page

Table 18 – continued from previous page

Industry Code	German Title	English Title
C2330000	Herstellung von keramischen Baumaterialien	Manufacture of ceramic building materials
C2340000	Herstellung von sonstiger Keramik	Manufacture of other ceramics
C2341000	Herstellung von keramischen Haushaltswaren	Manufacture of ceramic household articles
C2350000	Herstellung von Zement Kalk und gebranntem Gips	Manufacture of cement, lime and plaster
C2351000	Herstellung von Zement	Manufacture of cement
C2360000	Herstellung von Erzeugnissen aus Beton Zement und Gips	Manufacture of articles of concrete, cement and plaster
C2361000	Herstellung von Erzeugnissen aus Beton	Manufacture of concrete products
C2370000	Be- und Verarbeitung von Naturwerk- und Natursteinen	Cutting, shaping and finishing of stone
C2390000	Herstellung sonstiger nichtmetallischer Erzeugnisse a.n.g.	Manufacture of other non-metallic mineral products n.e.c.
C2391000	Herstellung von Schleifkörpern und Schleifmitteln	Manufacture of abrasive products
C2400000	Metallerzeugung und -bearbeitung	Manufacture of basic metals
C2420000	Herstellung von Stahlrohren	Manufacture of steel tubes
C2440000	Erzeugung und erste Bearbeitung von NE-Metallen	Manufacture of basic precious and other non-ferrous metals
C2450000	Gießereien	Foundries
C2450010	Eisen- Stahl- und Temperguss	Iron, steel and malleable cast iron
C2450020	NE-Metallguss	Non-ferrous metal casting
C2500000	Herstellung von Metallerzeugnissen	Manufacture of fabricated metal products
C2510000	Stahl- und Leichtmetallbau	Manufacture of metal structures
C2511000	Herstellung von Metallkonstruktionen	Manufacture of metal structures
C2512000	Herstellung von Ausbauelementen aus Metall	Manufacture of builders' carpentry and joinery of metal
C2520000	Herstellung von Metalltanks Heizkörpern und -kesseln	Manufacture of tanks, radiators and boilers

Continued on next page

Table 18 – continued from previous page

Industry Code	German Title	English Title
C2521000	Herstellung von Heizkesseln und -körpern für Zentralheizungen	Manufacture of central heating boilers and radiators
C2521120	Herstellung von Heizkesseln für Zentralheizungen	Manufacture of central heating boilers
C2529000	Herstellung von Behältern	Manufacture of containers
C2550000	Herstellung von Schmiede- Press- Zieh- und Stanzteilen	Forging, pressing, stamping and roll-forming of metal
C2560000	Oberflächenveredlung Wärmebehandlung und Mechanik a.n.g.	Surface treatment and coating of metals; general mechanical engineering
C2562000	Mechanik a.n.g.	General mechanical engineering
C2570000	Herstellung von Schneidwaren Werkzeugen Schlössern	Manufacture of cutlery, tools and general hardware
C2571000	Herstellung von Schneidwaren und Bestecken	Manufacture of cutlery
C2572000	Herstellung von Schlössern und Beschlägen	Manufacture of locks and hinges
C2573000	Herstellung von Werkzeugen	Manufacture of tools
C2573010	Herstellung von Werkzeugen (ohne Präzisionswerkzeuge)	Manufacture of tools (except precision tools)
C2573020	Herstellung von Präzisionswerkzeugen	Manufacture of precision tools
C2590000	Herstellung von sonstigen Metallwaren	Manufacture of other fabricated metal products
C2592000	Herstellung von Verpackungen und Verschlüssen aus Metall	Manufacture of light metal packaging
C2594000	Herstellung von Schrauben und Nieten	Manufacture of screws and rivets
C2600000	Datenverarbeitungsgeräte elektronische und optische Erzeugnisse	Computer, electronic and optical products
C2610000	Herstellung elektronischer Bauelemente und Leiterplatten	Manufacture of electronic components and boards
C2620000	Herstellung von DV-Geräten und peripheren Geräten	Manufacture of computers and peripheral equipment

Continued on next page

Table 18 – continued from previous page

Industry Code	German Title	English Title
C2630000	Herstellung von Geräten der Telekommunikationstechnik	Manufacture of communication equipment
C2651000	Herstellung von Mess- und Navigationsinstrumenten	Manufacture of measuring and navigation instruments
C2651100	Herstellung elektrischer Mess-Kontroll- Navigationsinstrumente	Manufacture of electrical measuring, testing, and navigation instruments
C2651200	Herstellung feinmechanisch-optischer Messinstrumente	Manufacture of precision mechanical and optical measuring instruments
C2670000	Herstellung von optischen und fotografischen Geräten	Manufacture of optical and photographic equipment
C2670010	Herstellung von optischen Instrumenten und Lasern	Manufacture of optical instruments and lasers
C2700000	Herstellung von elektrischen Ausrüstungen	Manufacture of electrical equipment
C2710000	Herstellung von Elektromotoren Generatoren Transformatoren usw.	Manufacture of electric motors, generators, transformers, etc.
C2711000	Herstellung von Elektromotoren Generatoren Transformatoren	Manufacture of electric motors, generators, transformers
C2712000	Elektrizitätsverteilungs- und Schalteinrichtungen	Electricity distribution and control apparatus
C2730000	Herstellung von Kabeln und Elektroinstallationsmaterial	Manufacture of cables and wiring devices
C2732000	Herstellung isolierter Elektrokabel -leitungen und -drähte	Manufacture of other electronic and electric wires and cables
C2733000	Herstellung von elektrischem Installationsmaterial	Manufacture of electrical installation material
C2740000	Herstellung von elektrischen Lampen und Leuchten	Manufacture of electric lamps and lighting equipment
C2750000	Herstellung von Haushaltsgeräten	Manufacture of domestic appliances
C2800000	Maschinenbau	Mechanical engineering
C2810000	Herstellung nicht wirtschaftszweigspezifischer Maschinen	Manufacture of general-purpose machinery

Continued on next page

Table 18 – continued from previous page

Industry Code	German Title	English Title
C2812000	Herstellung von hydraulischen und pneumatischen Komponenten	Manufacture of hydraulic and pneumatic components
C2812110	Herstellung von Ölhydraulik	Manufacture of oil hydraulics
C2812210	Herstellung von pneumatischen Steuerungen	Manufacture of pneumatic controls
C2813010	Herstellung von Flüssigkeitspumpen	Manufacture of liquid pumps
C2814000	Herstellung von Armaturen a.n.g.	Manufacture of other taps and valves
C2815000	Herstellung von Lagern Getrieben Zahnradern	Manufacture of bearings, gears, gearing and driving elements
C2815100	Herstellung von Lagern	Manufacture of bearings
C2815200	Herstellung von Getrieben Zahnradern und Antriebselementen	Manufacture of gears, gearing and driving elements
C2820000	Herstellung von nicht-branchenspezifischen Maschinen	Manufacture of non-industry specific machinery
C2822000	Herstellung von Hebezeugen und Fördermitteln	Manufacture of lifting and handling equipment
C2825000	Herstellung von kälte- und luft-technischen Erzeugnissen	Manufacture of refrigerating and air conditioning equipment
C2829000	Herstellung sonstiger nicht spezifischer Maschinen a.n.g.	Manufacture of other general-purpose machinery n.e.c.
C2830000	Herstellung von land- und forstwirtschaftlichen Maschinen	Manufacture of agricultural and forestry machinery
C2840000	Herstellung von Werkzeugmaschinen	Manufacture of machine tools
CONG0CN0	Herstellung von Verbrauchsgütern (ohne Ernährungsgewerbe)	Manufacture of consumer goods (excluding food)
C00FOBE0	Ernährungsgewerbe und Tabakverarbeitung	Food and tobacco industry
C1000000	Herstellung von Nahrungs- und Futtermitteln	Manufacture of food and feed products

Continued on next page

Table 18 – continued from previous page

Industry Code	German Title	English Title
C1010000	Schlachten und Fleischverarbeitung	Slaughtering and meat processing
C1020000	Fischverarbeitung	Fish processing
C1030000	Obst- und Gemüseverarbeitung	Fruit and vegetable processing
C1050000	Milchverarbeitung	Dairy processing
C1060000	Herstellung von Mahl- und Schälmaschinen	Manufacture of grain mill products, starches and starch products
C2841000	Herstellung von Werkzeugmaschinen für die Metallbearbeitung	Manufacture of machine tools for metalworking
C2849200	Herstellung von Maschinen zur Bearbeitung harter Stoffe	Manufacture of machines for processing hard materials
C2890000	Herstellung von Maschinen für sonstige bestimmte Branchen	Manufacture of machinery for other specific industries
C2891000	Herstellung von Metallformmaschinen	Manufacture of metal forming machinery
C2892000	Herstellung von Bergwerks- Bau- und Baustoffmaschinen	Manufacture of mining, quarrying and construction machinery
C2892200	Herstellung von Bau- und Baustoffmaschinen	Manufacture of building and building material machines
C2893000	Herstellung von Maschinen für das Ernährungsgewerbe	Manufacture of machinery for food, beverage and tobacco processing
C2894000	Herstellung von Maschinen für das Textilgewerbe	Manufacture of machinery for textile, apparel and leather production
C2896000	Herstellung von Gummi- und Kunststoffindustriemaschinen	Manufacture of rubber and plastics machinery
C2899000	Herstellung sonstiger branchenspezifischer Maschinen a.n.g.	Manufacture of other special-purpose machinery n.e.c.
C2900000	Herstellung von Kraftwagen und Kraftwagenteilen	Manufacture of motor vehicles, trailers and semi-trailers
C2910000	Herstellung von Kraftwagen und -motoren	Manufacture of motor vehicles and engines
C2911000	Herstellung von Nutzkraftwagen	Manufacture of commercial vehicles

Continued on next page

Table 18 – continued from previous page

Industry Code	German Title	English Title
C2912000	Herstellung von Personenkraftwagen einschließlich Kombis	Manufacture of passenger cars including station wagons
C2920000	Herstellung von Karosserien Aufbauten und Anhängern	Manufacture of bodies (coachwork) for motor vehicles; manufacture of trailers and semi-trailers
C2930000	Herstellung von Teilen und Zubehör für Kraftwagen	Manufacture of parts and accessories for motor vehicles
C3000000	Sonstiger Fahrzeugbau	Other vehicle construction
C3100000	Herstellung von Möbeln	Manufacture of furniture
C3101000	Herstellung von Büro- und Ladenmöbeln	Manufacture of office and shop furniture
C3109000	Herstellung von sonstigen Möbel	Manufacture of other furniture
C3109040	Herstellung von Polstermöbeln	Manufacture of upholstered furniture
C3200000	Herstellung von sonstigen Waren	Manufacture of other goods
C3250000	Herstellung von medizinischen Apparaten und Materialien	Manufacture of medical and dental instruments and supplies
C3250200	Herstellung von medizinmechanischen Erzeugnissen	Manufacture of medical mechanical products
C3291000	Herstellung von Besen und Bürsten	Manufacture of brooms and brushes
C1070000	Herstellung von Back- und Teigwaren	Manufacture of bread, fresh pastry goods and cakes
C1071000	Herstellung von Backwaren (ohne Dauerbackwaren)	Manufacture of bread; manufacture of fresh pastry goods and cakes
C1080000	Herstellung von sonstigen Nahrungsmitteln	Manufacture of other food products
C1082000	Herstellung von Süßwaren (einschließlich Dauerbackwaren)	Manufacture of cocoa, chocolate and sugar confectionery
C1090000	Herstellung von Futtermitteln	Manufacture of prepared animal feeds
C1100000	Getränkeherstellung	Manufacture of beverages
C1101000	Herstellung von Spirituosen	Manufacture of spirits
C1105000	Herstellung von Bier	Manufacture of beer

Continued on next page

Table 18 – continued from previous page

Industry Code	German Title	English Title
C1107000	Herstellung von Erfrischungsgetränken und Mineralwasser	Manufacture of soft drinks; production of mineral waters and other bottled waters
C1300000	Herstellung von Textilien	Manufacture of textiles
C1320000	Weberei	Weaving of textiles
C1321000	Baumwollweberei	Cotton weaving
C1330000	Veredlung von Textilien und Bekleidung	Finishing of textiles
C1390000	Herstellung von sonstigen Textilwaren	Manufacture of other textiles
C1392000	Herstellung konfektionierter Textilwaren (ohne Bekleidung)	Manufacture of made-up textile articles, except apparel
C1400000	Herstellung von Bekleidung	Manufacture of wearing apparel
C1410000	Herstellung von Bekleidung (ohne Pelzbekleidung)	Manufacture of wearing apparel, except fur apparel
C1413000	Herstellung von sonstiger Oberbekleidung	Manufacture of other outerwear
C1414000	Herstellung von Wäsche	Manufacture of underwear
C1414100	Herstellung von gewebter Wäsche (ohne Miederwaren)	Manufacture of woven underwear (except corsetry)
C1431000	Herstellung von Strumpfwaren	Manufacture of knitted and crocheted hosiery
C1500000	Herstellung von Leder Lederwaren und Schuhen	Manufacture of leather and related products
C1510000	Herstellung von Leder und Lederwaren (ohne Lederbekleidung)	Tanning and dressing of leather; manufacture of luggage, handbags, saddlery and harness
C1520000	Herstellung von Schuhen	Manufacture of footwear
C1600000	Holz- Flecht- Korb- und Korkwarenherstellung (ohne Möbel)	Manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials
C1610000	Säge- Hobel- und Holzimprägnierwerke	Sawmilling and planing of wood

Continued on next page

Table 18 – continued from previous page

Industry Code	German Title	English Title
C1620000	Herstellung sonstiger Holz-Flecht- Korb- und Korkwaren	Manufacture of other products of wood; manufacture of articles of cork, straw and plaiting materials
C1621000	Herstellung von Furnier-Sperrholz- und Holzspanplatten	Manufacture of veneer sheets and wood-based panels
C1623000	Herstellung von Fertigbauteilen und Ausbauelementen	Manufacture of builders' carpentry and joinery
C1624000	Herstellung von Verpackungsmitteln aus Holz	Manufacture of wooden containers
C1700000	Papiergewerbe	Paper industry
C1710000	Herstellung von Holz- und Zellstoff Papier Karton und Pappe	Manufacture of pulp, paper and paperboard
C1712000	Herstellung von Papier Karton und Pappe	Manufacture of paper and paperboard
C1712100	Herstellung von Papier	Manufacture of paper
C1712200	Herstellung von Karton und Pappe	Manufacture of paperboard
C1720000	Papier- Karton- und Pappeverarbeitung	Manufacture of articles of paper and paperboard
C1721000	Herstellung von Wellpapier und -pappe Verpackungsmitteln	Manufacture of corrugated paper and paperboard and of containers of paper and paperboard
C1723000	Herstellung von Schreibwaren und Bürobedarf aus Papier	Manufacture of paper stationery
C1800000	Herstellung von Druckerzeugnissen	Printing and service activities related to printing
C1810000	Herstellung von Druckerzeugnissen	Printing and service activities related to printing
C1811000	Zeitungsdruck	Newspaper printing
C1812000	Drucken a.n.g.	Other printing
C1813000	Druck und Medieneinstufung	Pre-press and pre-media services
C1900000	Kokerei und Mineralölverarbeitung	Coke and refined petroleum products manufacturing
C2000000	Herstellung von chemischen Erzeugnissen	Manufacture of chemicals and chemical products

Continued on next page

Table 18 – continued from previous page

Industry Code	German Title	English Title
C2010000	Herstellung von chemischen Grundstoffen	Manufacture of basic chemicals
C2012000	Herstellung von Farbstoffen und Pigmenten	Manufacture of dyes and pigments
C2014000	Herstellung sonstiger organischer Grundstoffe	Manufacture of other organic basic chemicals
C2016000	Herstellung von Kunststoff in Primärformen	Manufacture of plastics in primary forms
C2030000	Herstellung von Anstrichmitteln Druckfarben und Kitt	Manufacture of paints, varnishes and similar coatings, printing ink and mastics
C2040000	Herstellung von Wasch- Reinigungs- u. Körperpflegemitteln	Manufacture of soap and detergents, cleaning and polishing preparations, perfumes and toilet preparations
C2050000	Herstellung von sonstigen chemischen Erzeugnissen	Manufacture of other chemical products
C2100000	Herstellung von pharmazeutischen Erzeugnissen	Manufacture of pharmaceuticals, medicinal chemical and botanical products
C2200000	Herstellung von Gummi- und Kunststoffwaren	Manufacture of rubber and plastic products
C2210000	Herstellung von Gummiwaren	Manufacture of rubber products
C2219000	Herstellung von sonstigen Gummiwaren	Manufacture of other rubber products

9 Electronic Supplement

The source-code for this report and the statistical analysis can be found on Github at <https://github.com/dgloukhman/ifo-lmu-consulting>. The survey data is not public and can be obtained from the ifo Institute.